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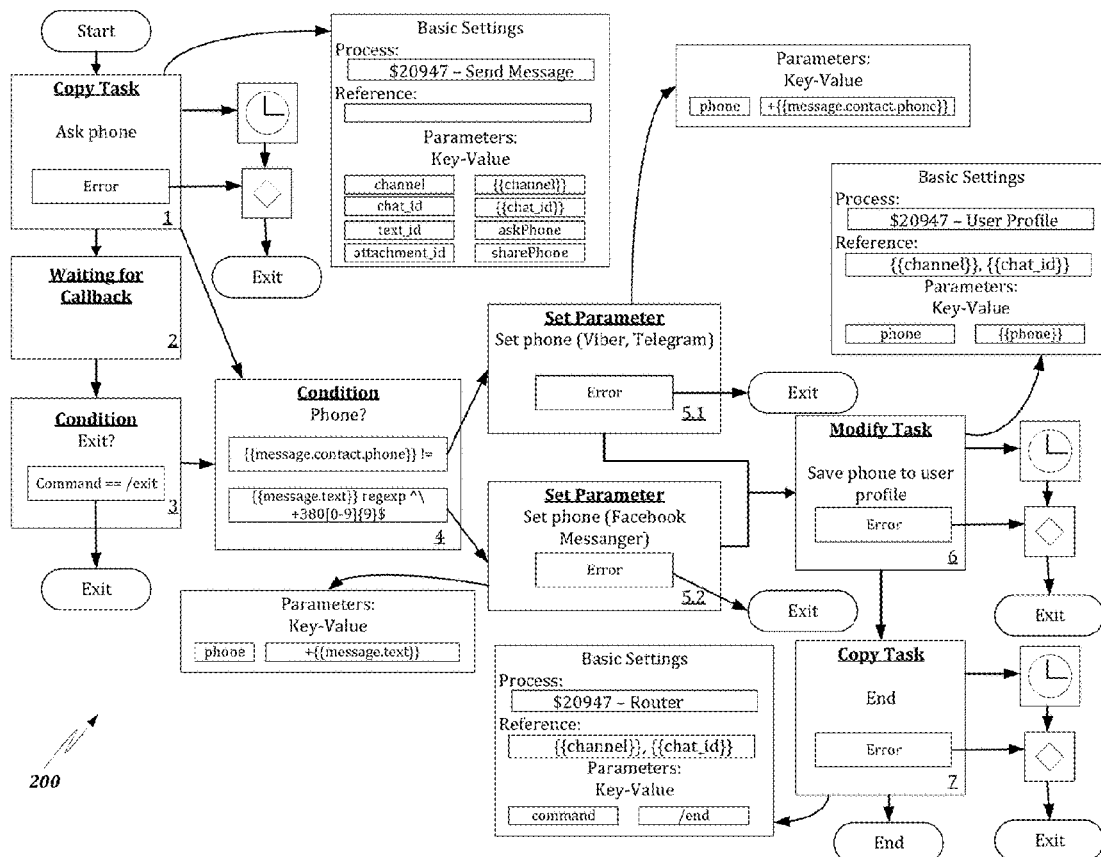
(19) **United States**(12) **Patent Application Publication**
Vityaz(10) **Pub. No.: US 2022/0100554 A1**(43) **Pub. Date: Mar. 31, 2022**(54) **SYSTEM AND METHOD FOR MODIFYING A DATA PROCESSING METHOD**(71) Applicant: **Middleware, Inc.**, Wilmington, DE (US)(72) Inventor: **Oleksandr Vityaz**, Dnipropetrovsk (UA)(21) Appl. No.: **17/481,742**(22) Filed: **Sep. 22, 2021****Related U.S. Application Data**

(63) Continuation-in-part of application No. 17/388,341, filed on Jul. 29, 2021, which is a continuation-in-part of application No. 16/252,075, filed on Jan. 18, 2019, now Pat. No. 11,237,835, which is a continuation-in-part of application No. 15/268,802, filed on Sep. 19, 2016, now abandoned, said application No. 16/252,075 is a continuation-in-part of application No. 15/077,626, filed on Mar. 22, 2016, now abandoned, said application No. 17/388,341 is a continuation-in-part of application No. 16/027,926, filed on Jul. 5, 2018, which is a continuation-in-part of application No. 15/268,802, filed on Sep. 19, 2016, now abandoned, said application No. 16/027,926 is a continuation-in-part of application No. 15/077,626, filed on Mar. 22, 2016, now abandoned.

(60) Provisional application No. 63/213,114, filed on Jun. 21, 2021, provisional application No. 62/221,124, filed on Sep. 21, 2015, now abandoned, provisional application No. 62/137,079, filed on Mar. 23, 2015, now abandoned, provisional application No. 62/221,124, filed on Sep. 21, 2015, now abandoned, provisional application No. 62/137,079, filed on Mar. 23, 2015, now abandoned.

Publication Classification(51) **Int. Cl.**
G06F 9/48 (2006.01)(52) **U.S. Cl.**
CPC **G06F 9/48** (2013.01)(57) **ABSTRACT**

Disclosed herein are systems and methods for multi-system connectivity and automation via universal computing elements. Universal computing elements may comprise an object queue, one or more counters, and a function operating on parameters of objects in the object queue. Universal computing elements may be interconnected into processes of arbitrary complexity. Universal computing elements may facilitate modular and scalable business process development, including application programming interface and database connectivity. The processes may be stored in a repository accessible to multiple entities, and may be modified prior to being integrated into a process flow of at least one of the entities.



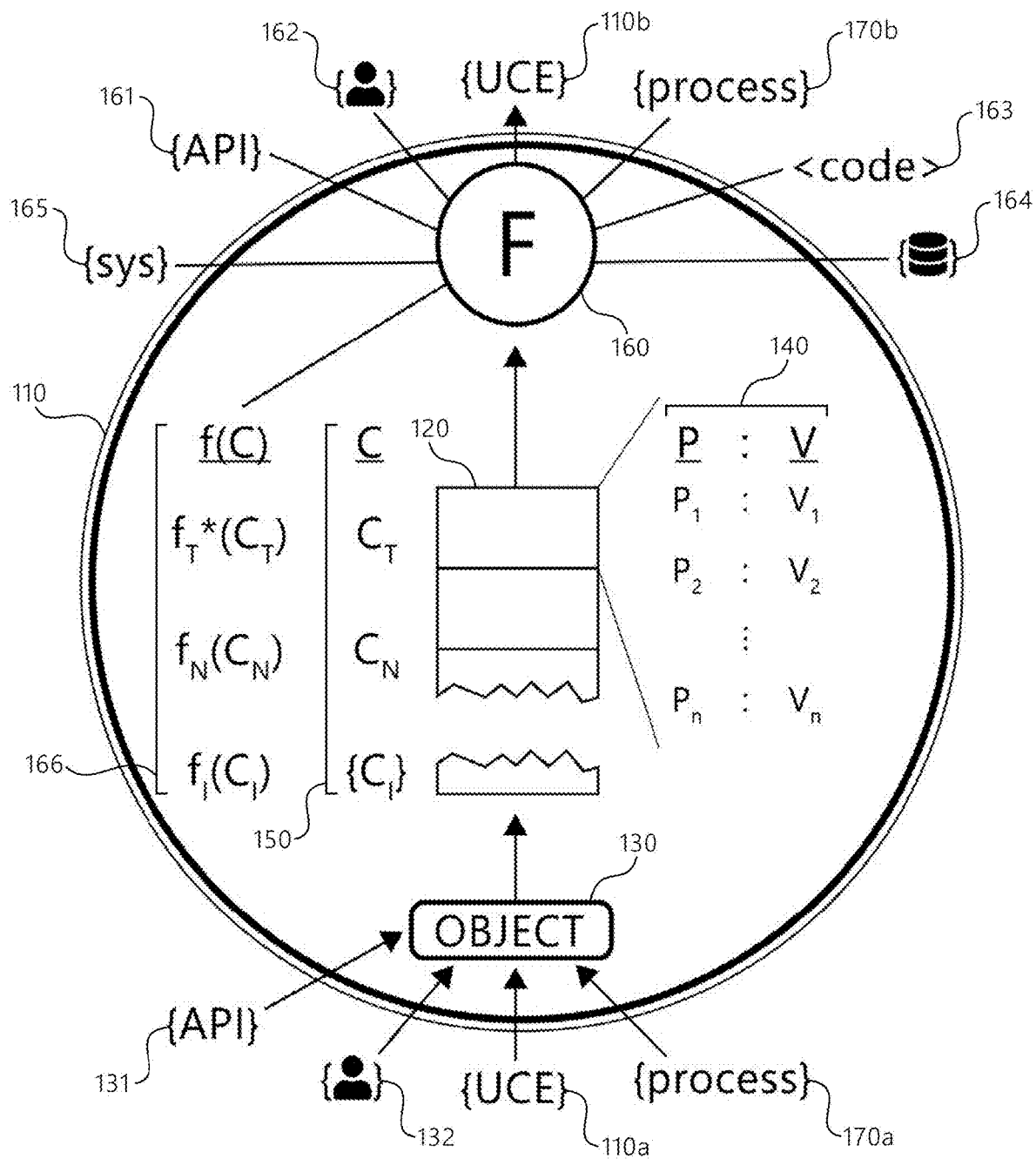
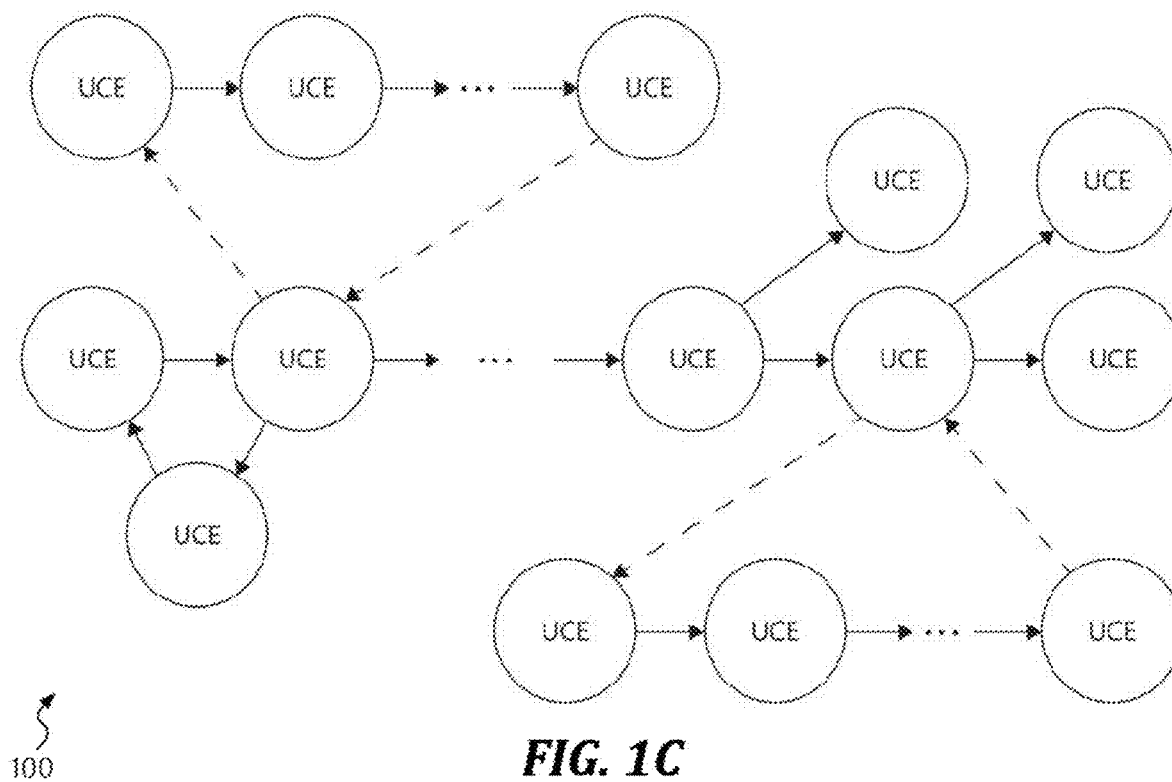
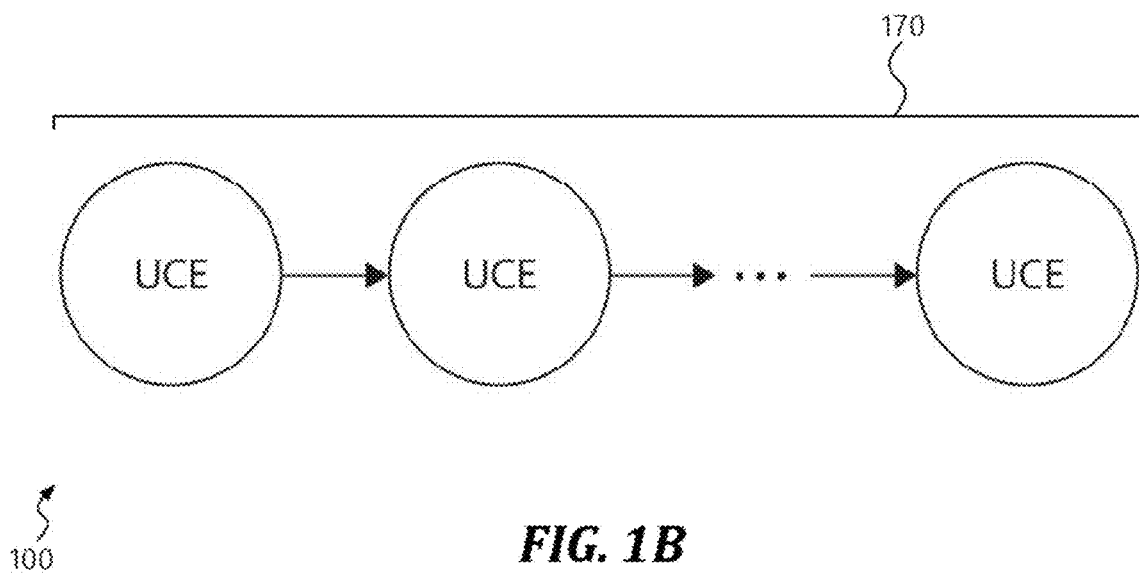


FIG. 1A



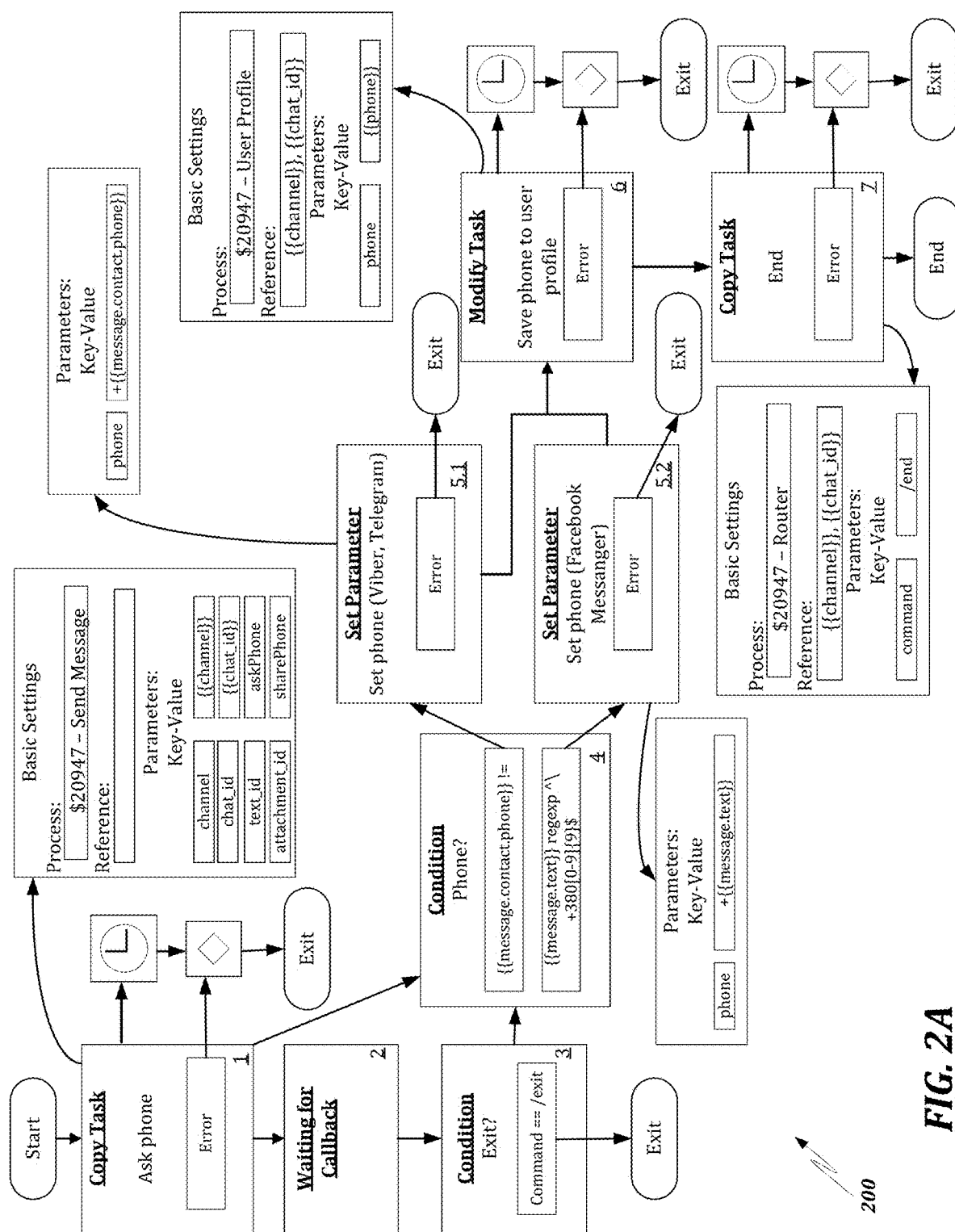


FIG. 2A

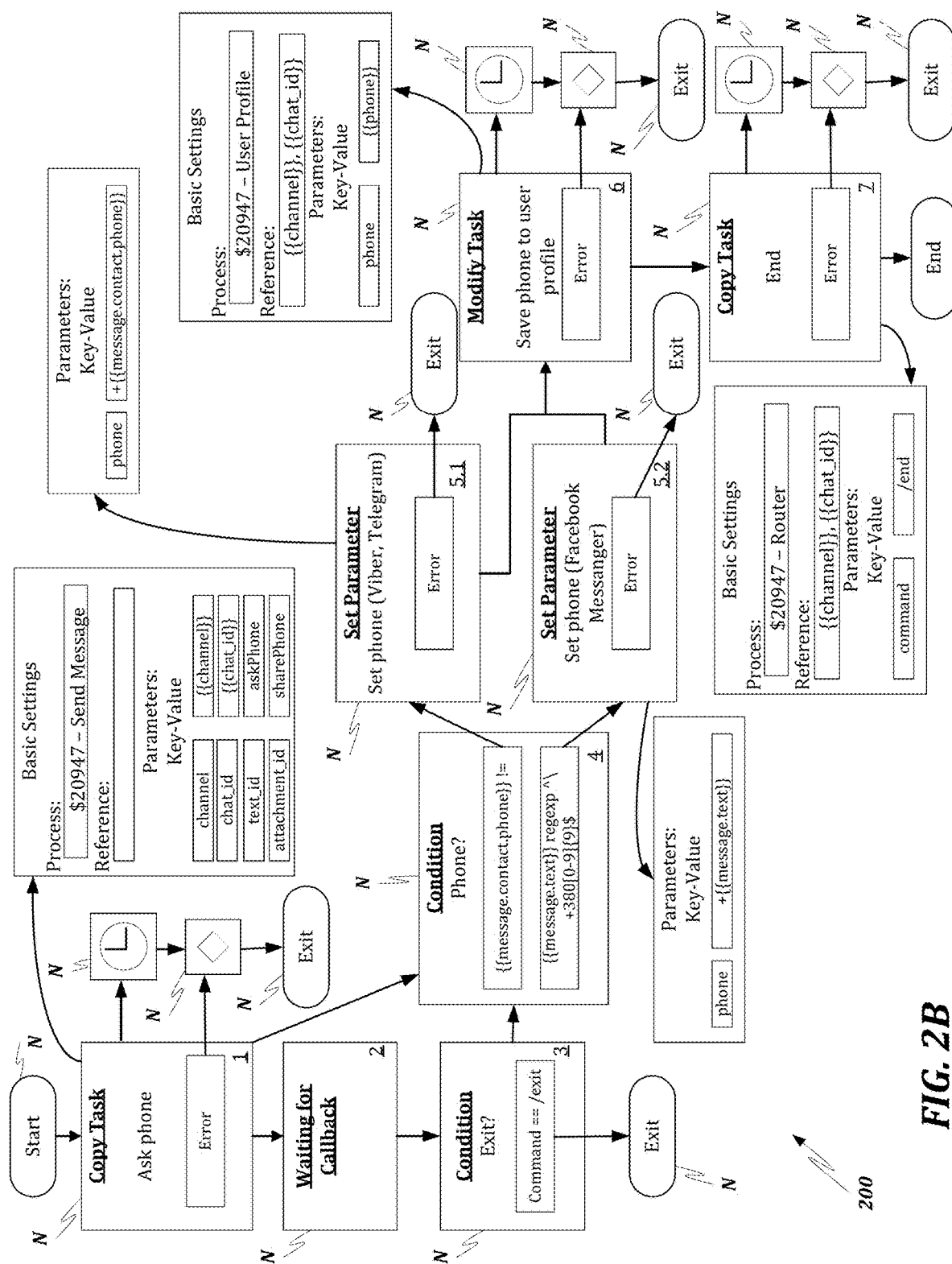


FIG. 2B

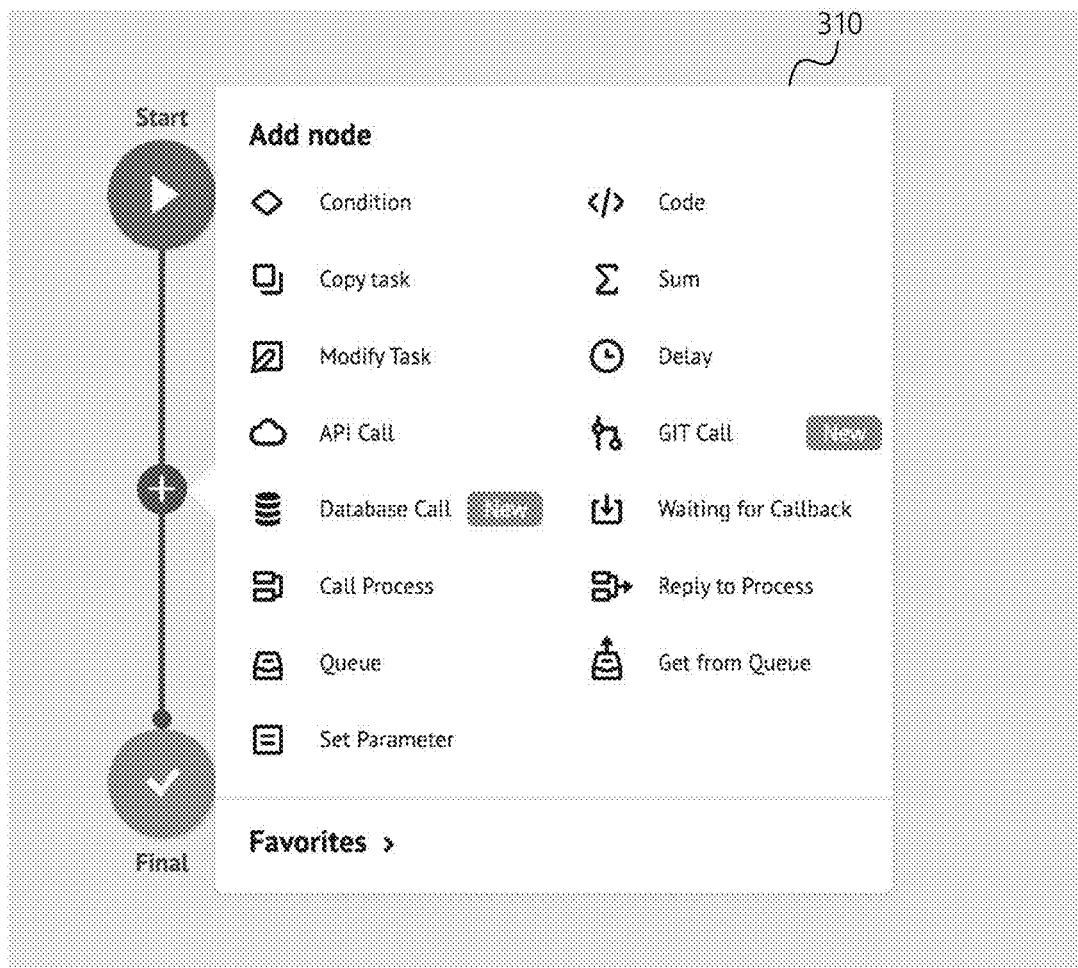


FIG. 3

300

API Call

Create access token

Add description

Show more

< URL API

https://www.googleapis.com/oauth2/v4/token

Request format: Default

Request method: POST

Content-Type:

Application/X-Web-Form-Urlencoded

< Parameters Code Editor

Key-Value	
redirect_uri	urn:ietf:wg:oauth:2.0...
grant_type	authorization_code
client_id	{{client_id}}
code	{{code}}
client_secret	{{client_secret}}

> Additionally

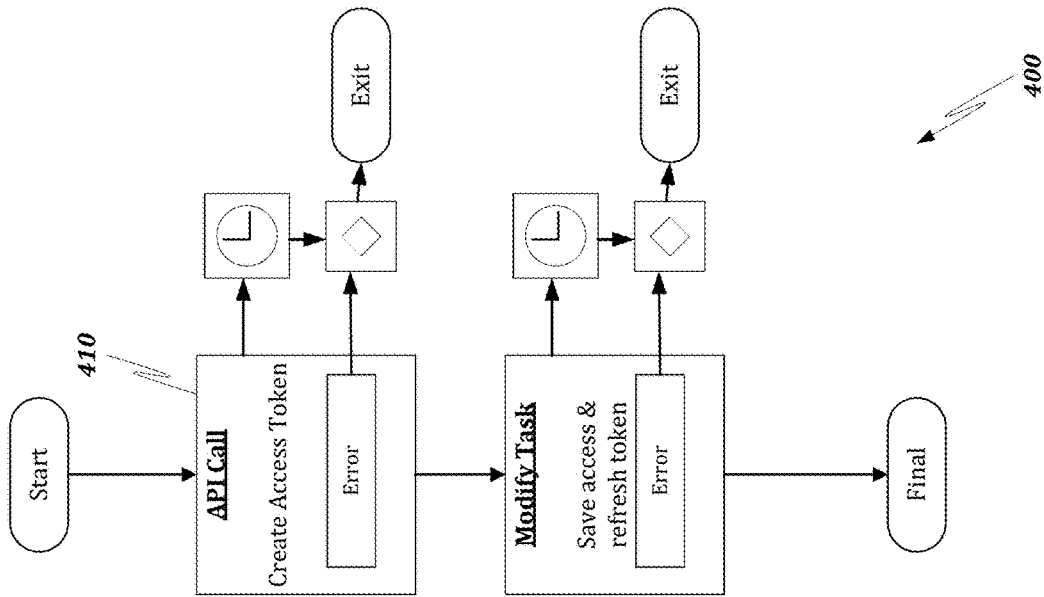
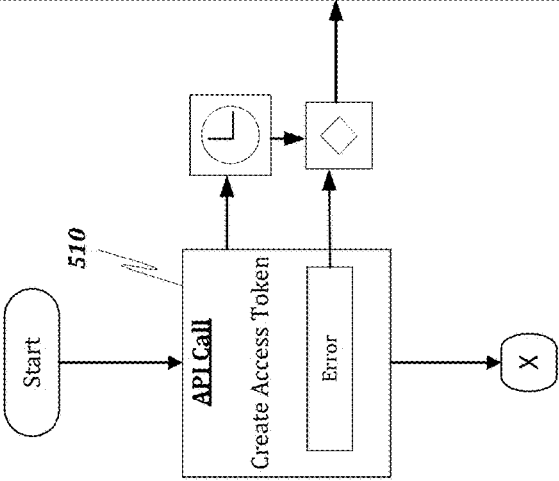
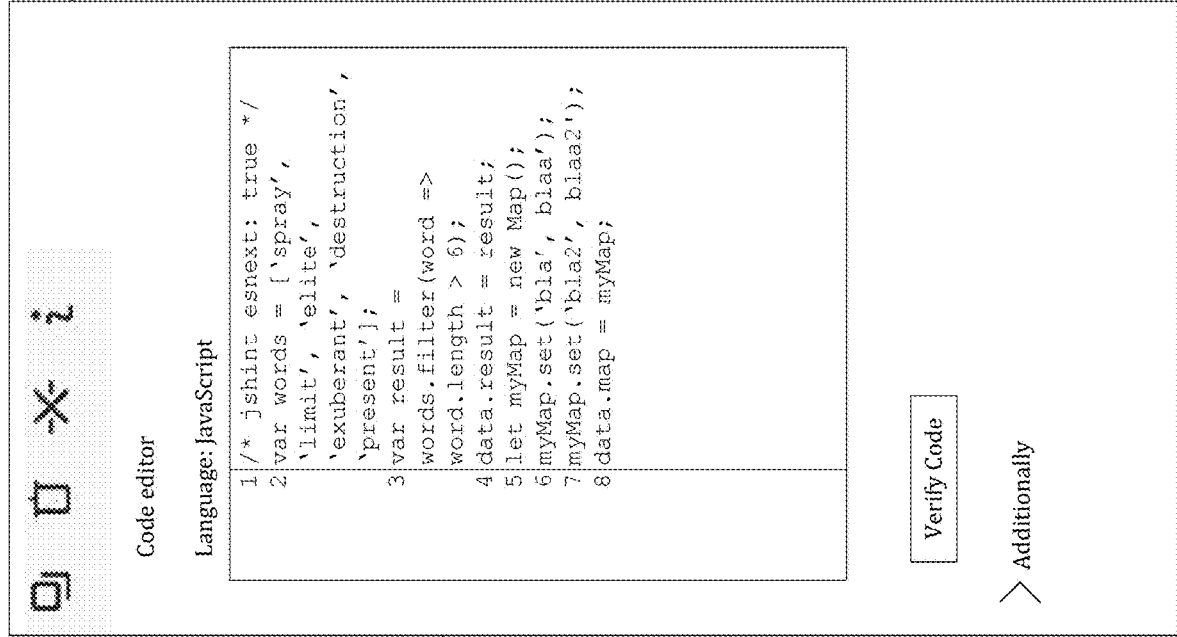


FIG. 4



500

FIG. 5

<>	Condition
</>	Code
	Copy task
Σ	Sum
	Modify Task
	Delay
	API Call
	GIT Call
	Waiting for Callback
	Call Process
	Reply to Process
	Queue
	Get from Queue
	Set Parameter

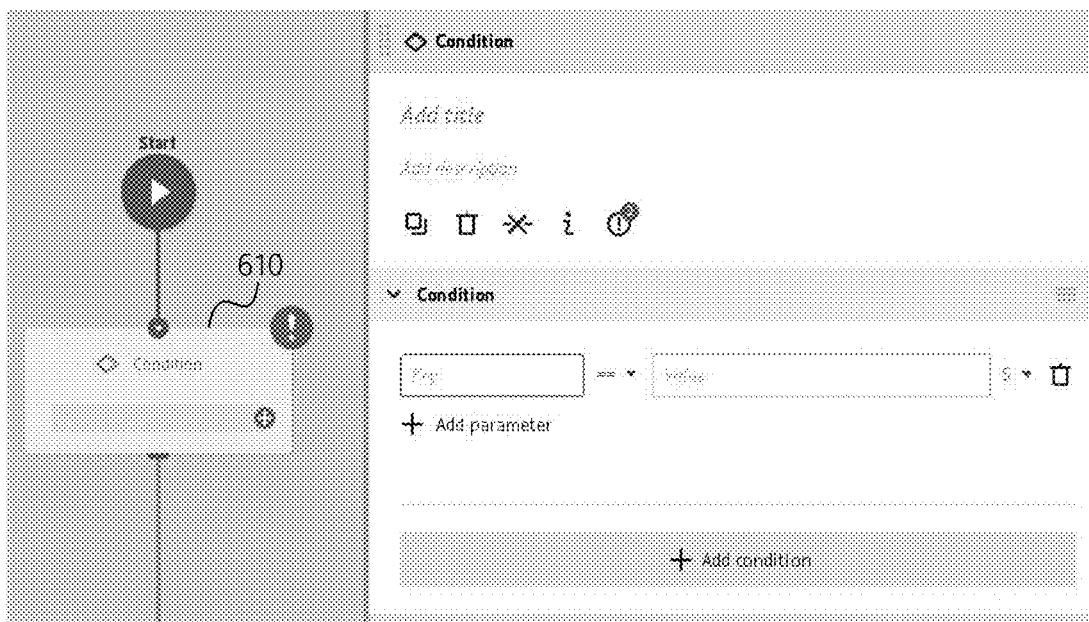


FIG. 6

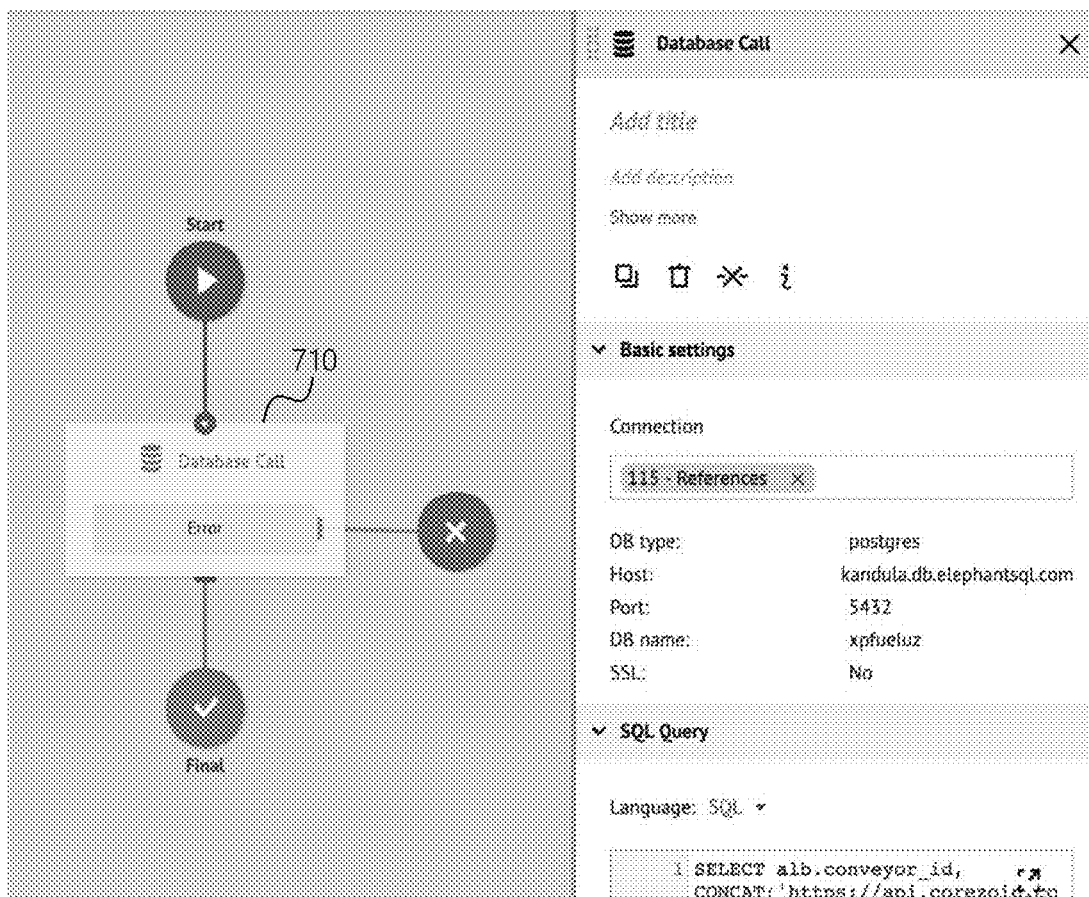


FIG. 7

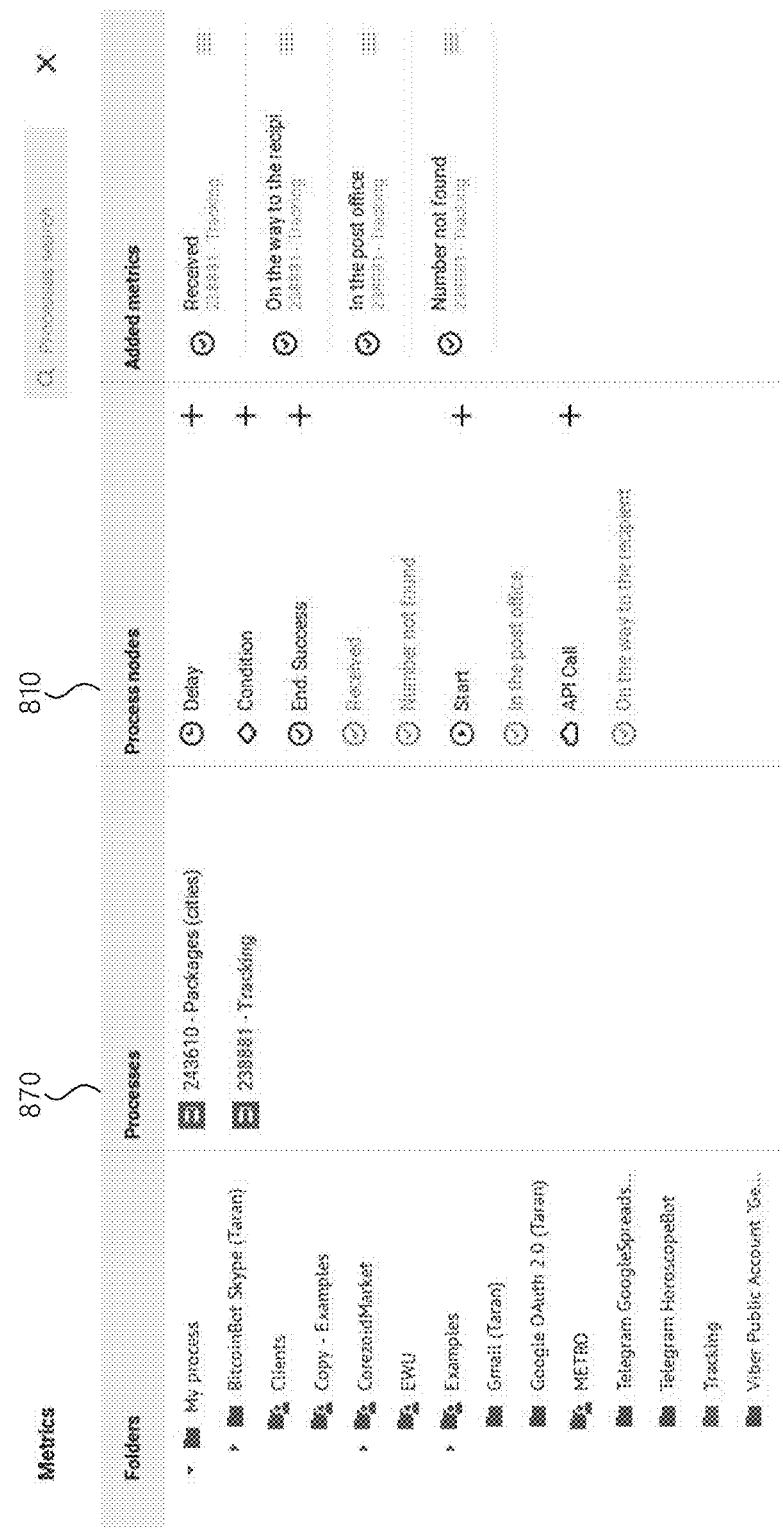


FIG. 8A

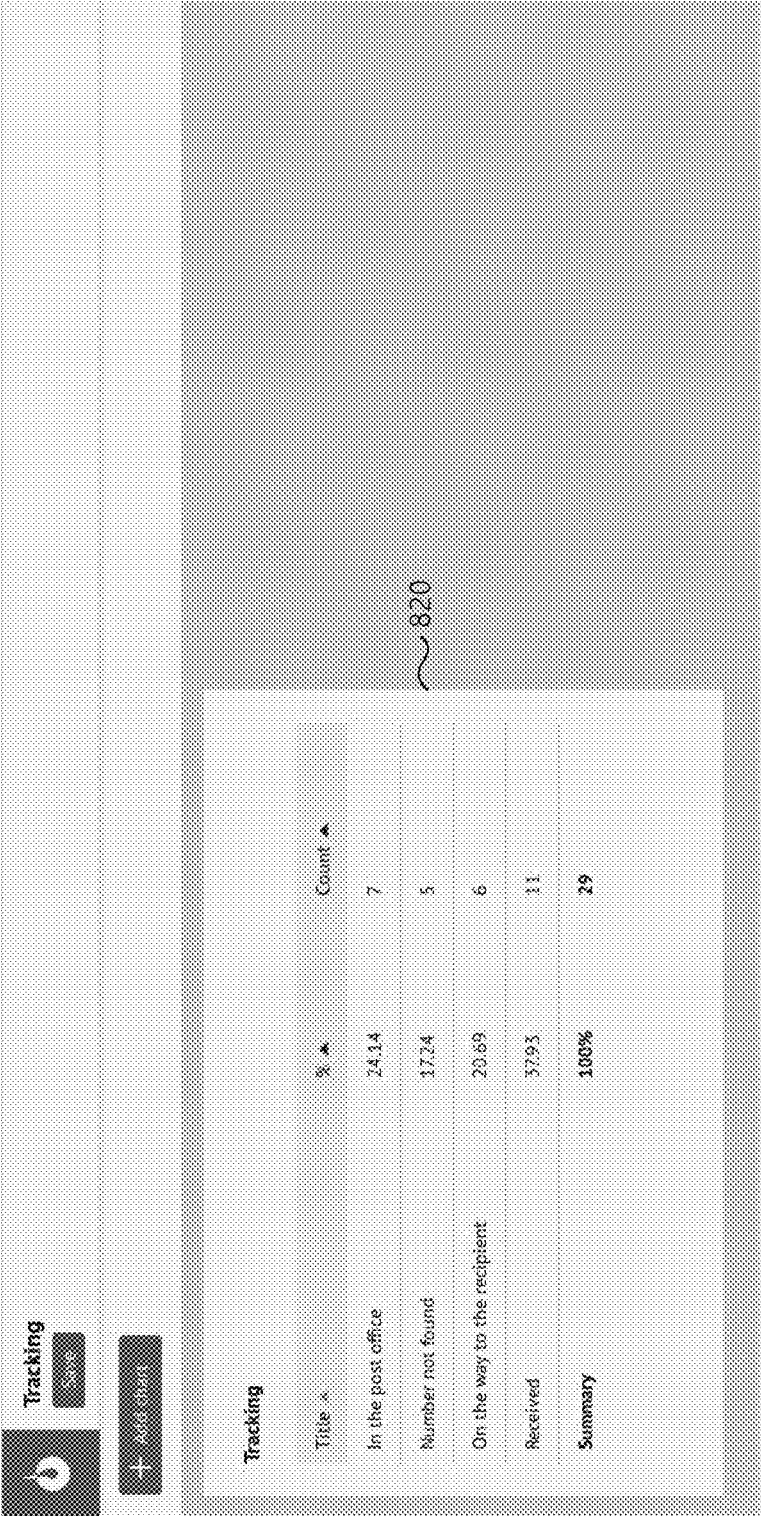


FIG. 8B

800

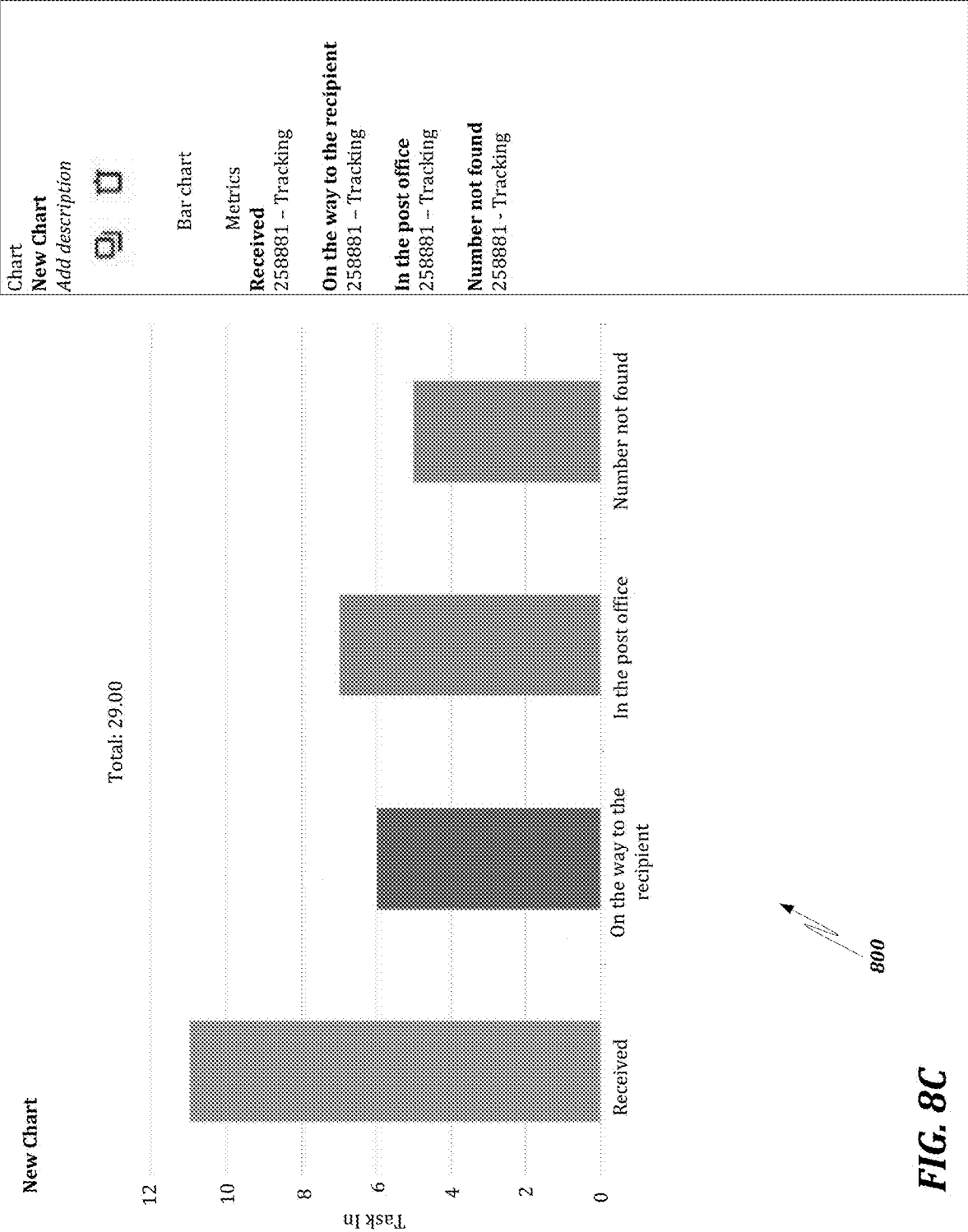


FIG. 8C

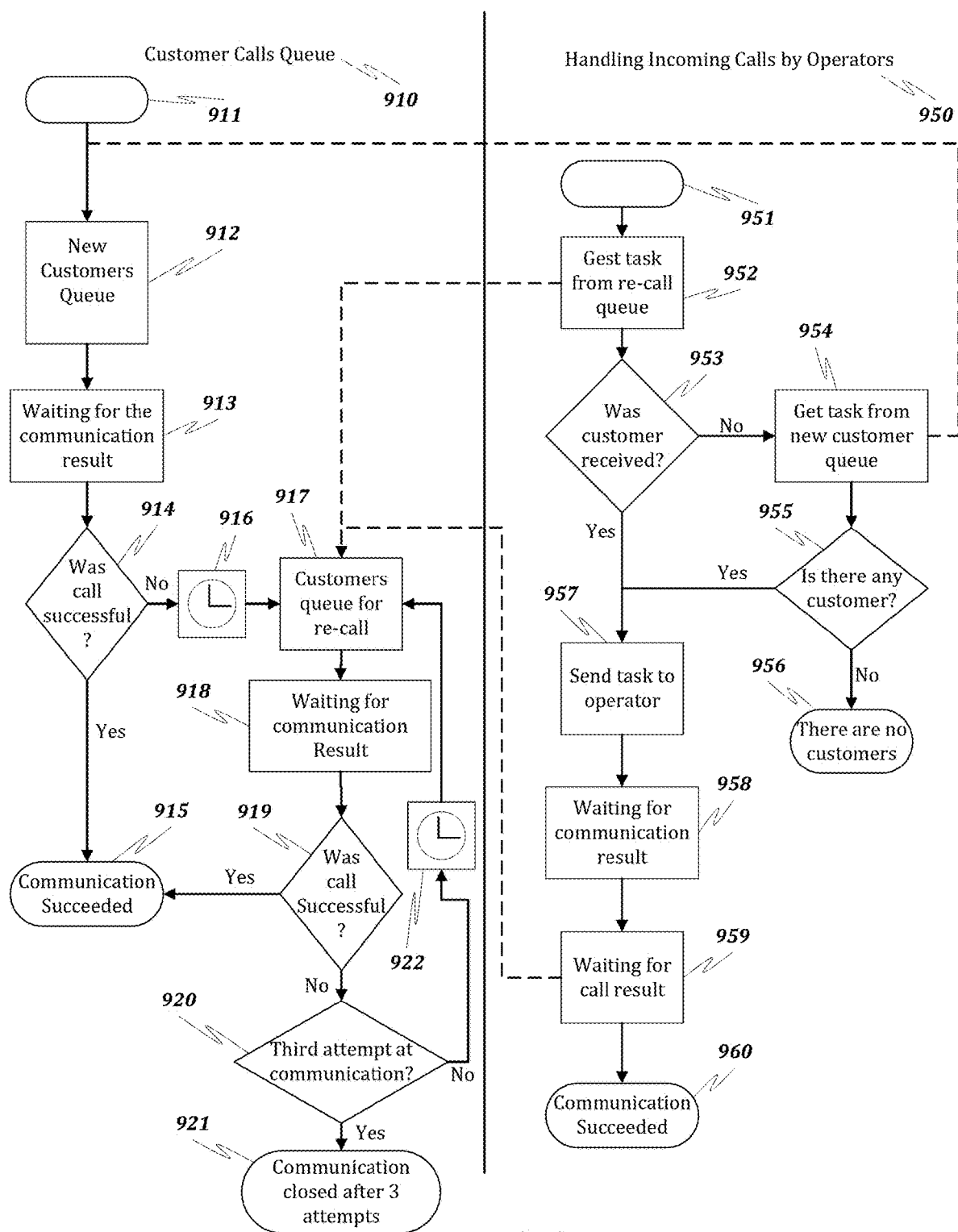


FIG. 9A

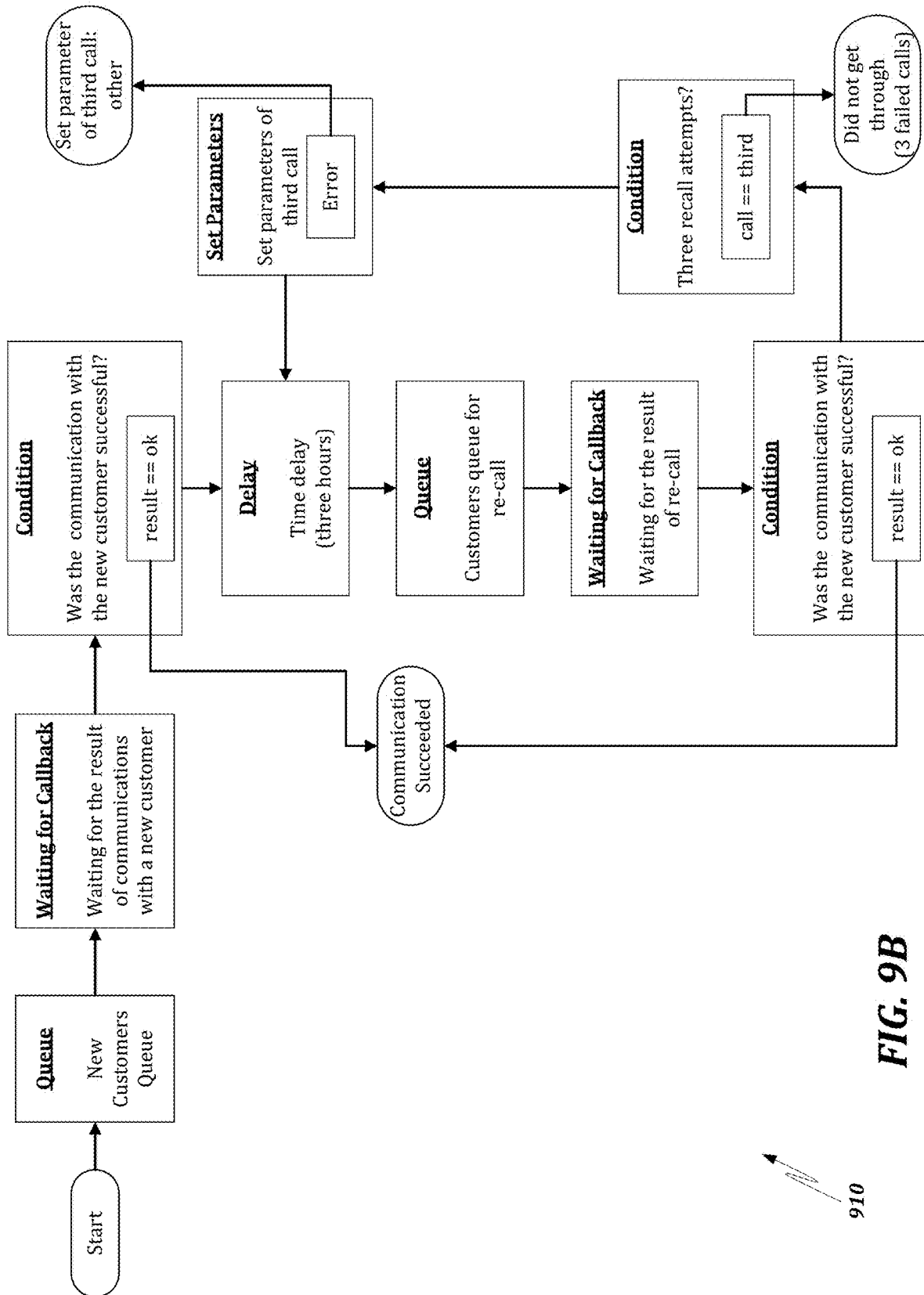


FIG. 9B

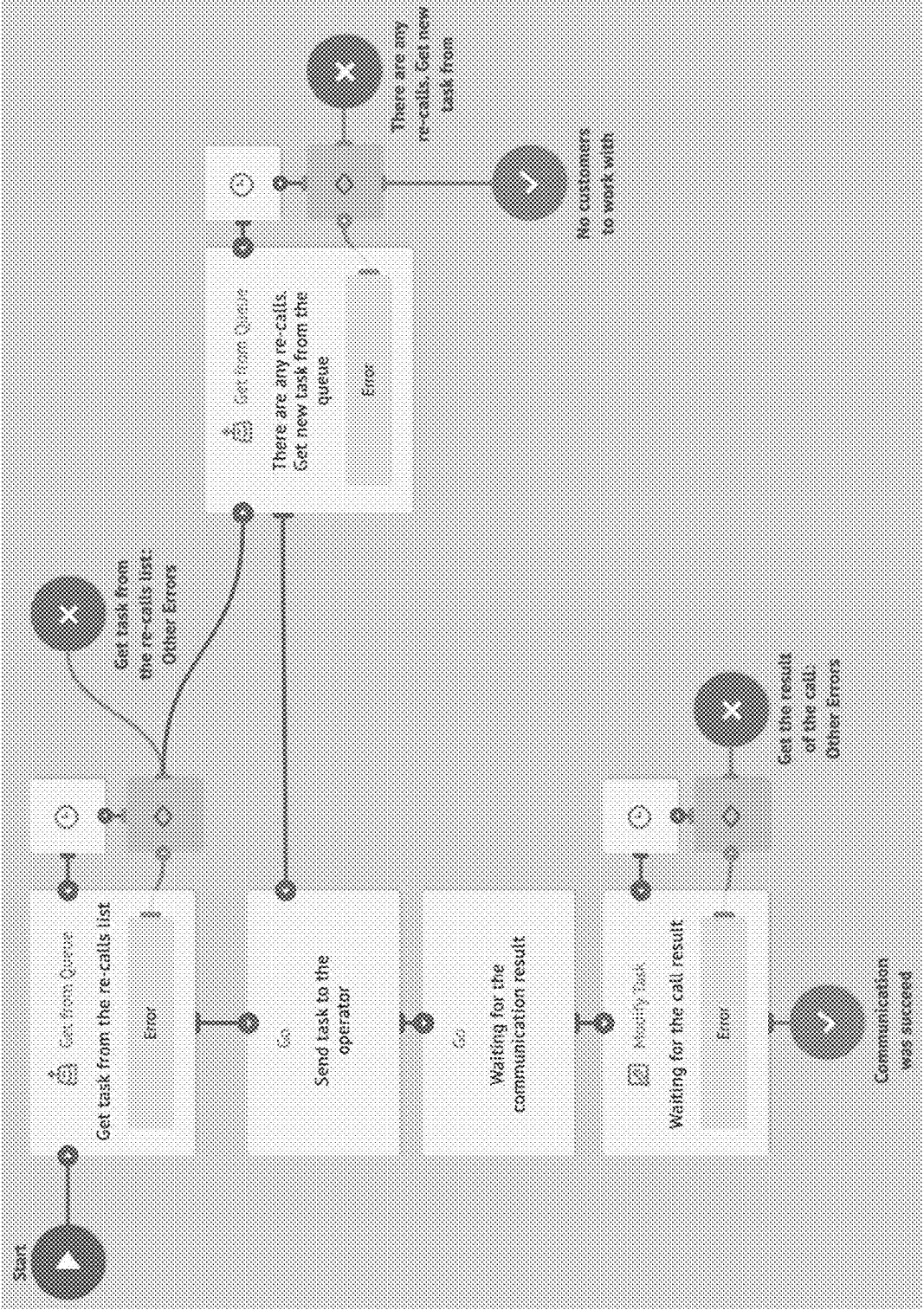
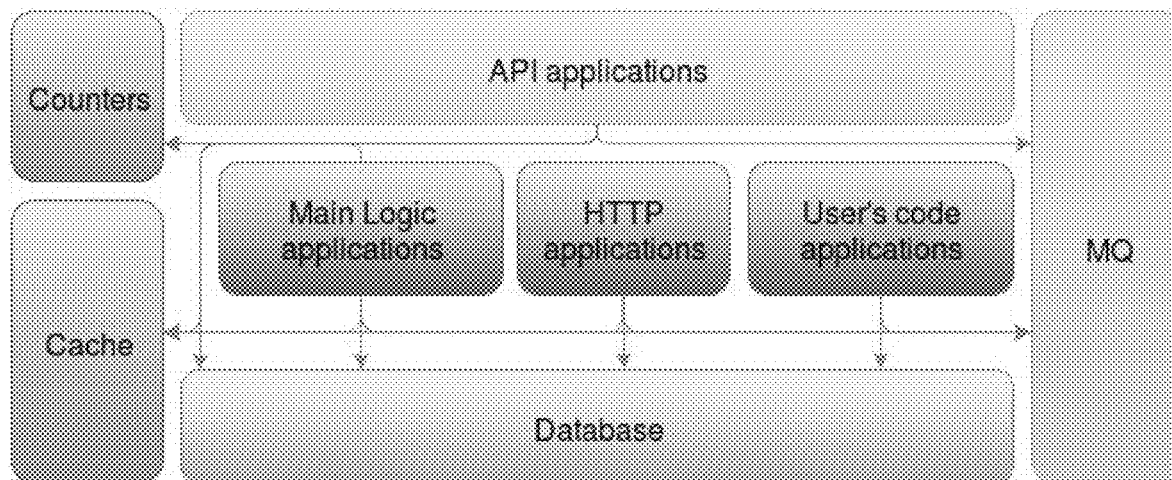
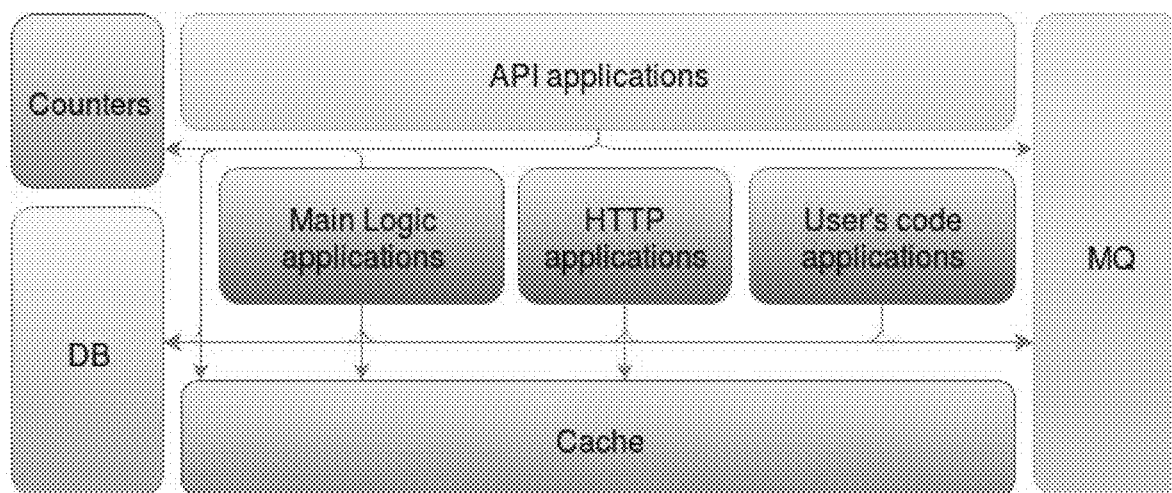


FIG. 9C



1000

FIG. 10



1100

FIG. 11

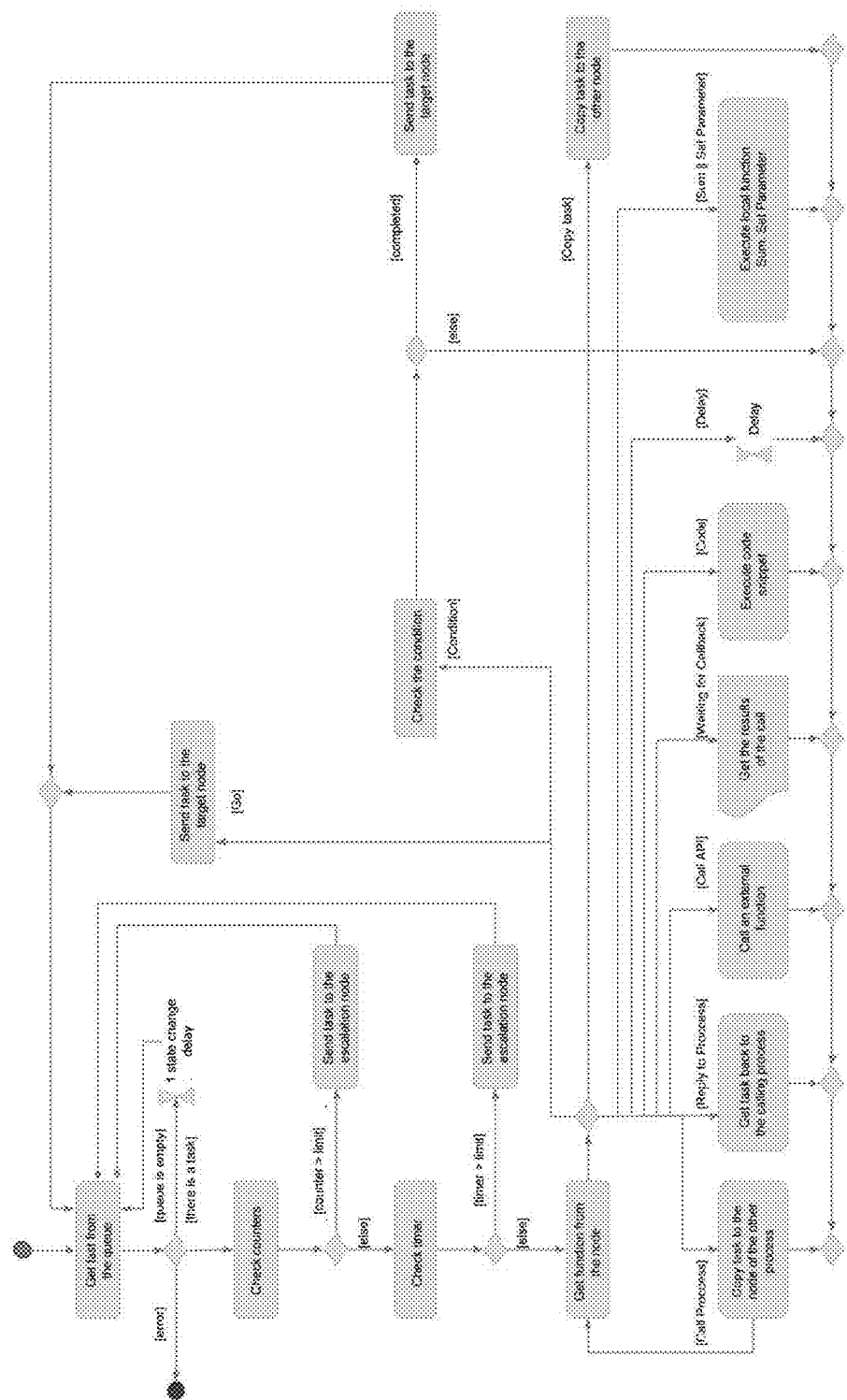


FIG. 12

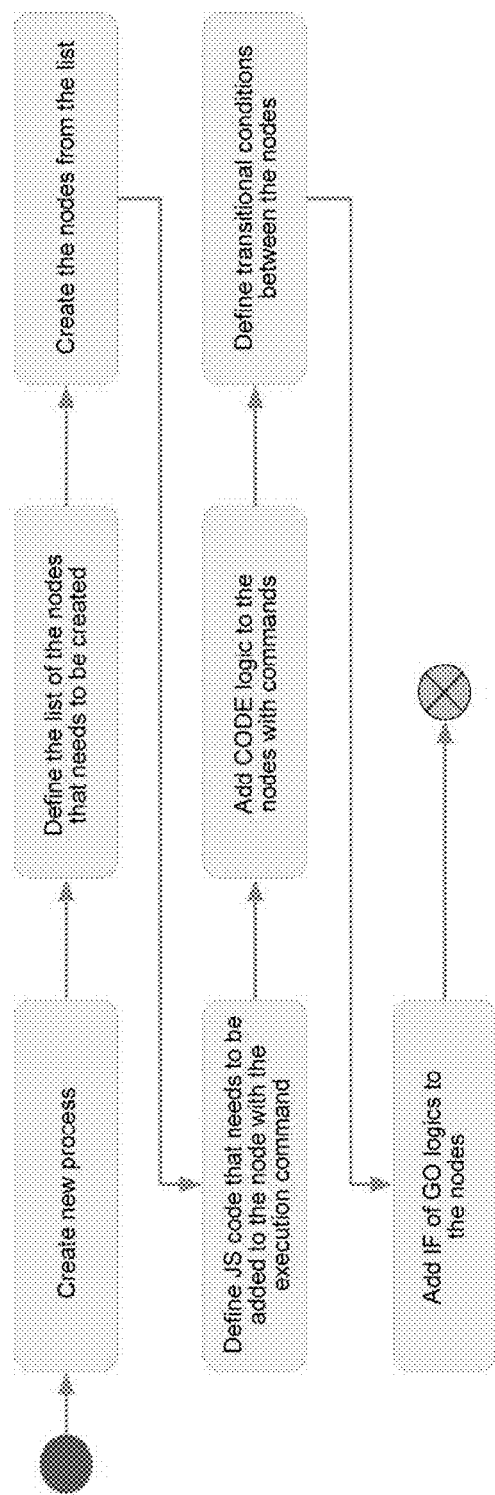
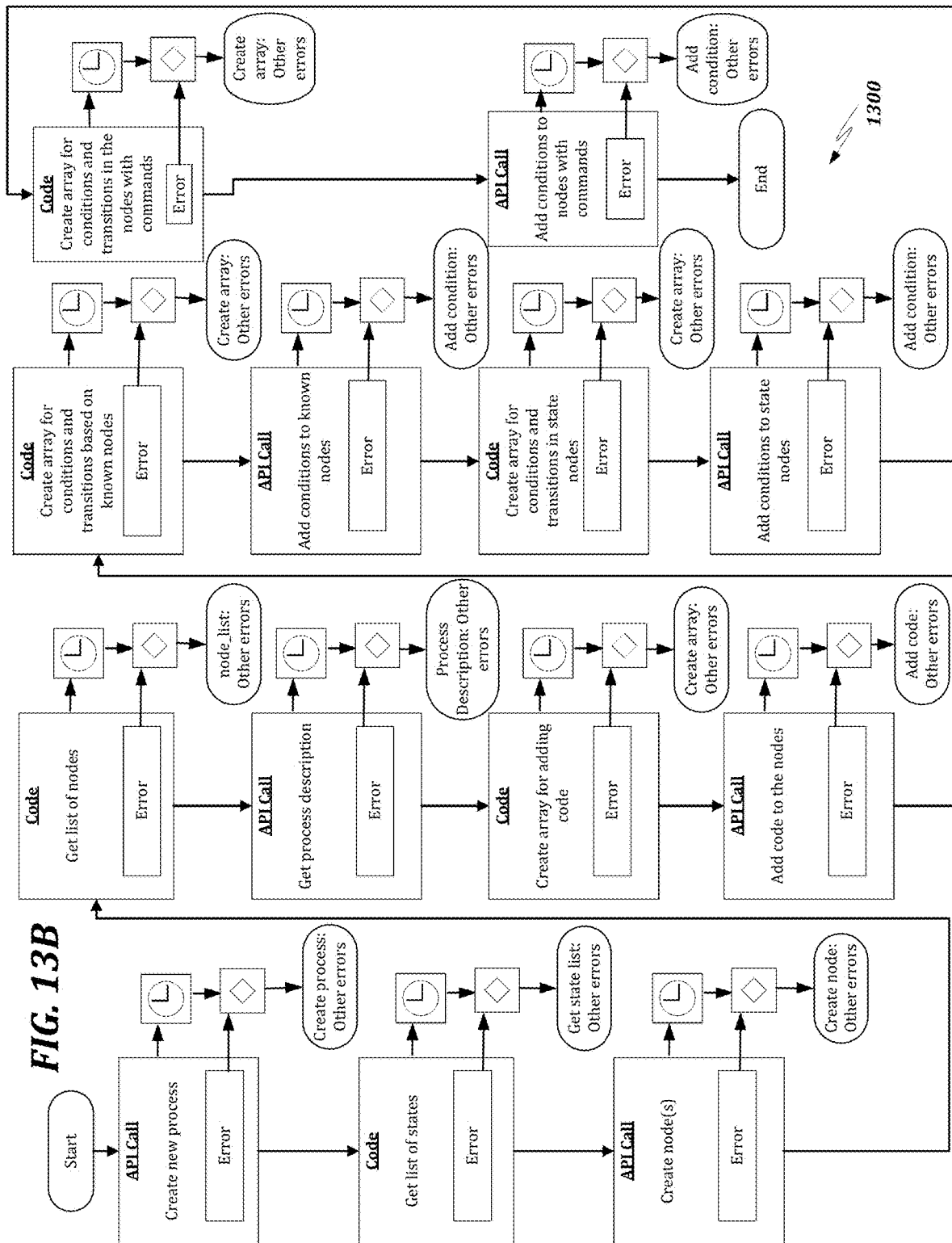


FIG. 13A

FIG. 13B



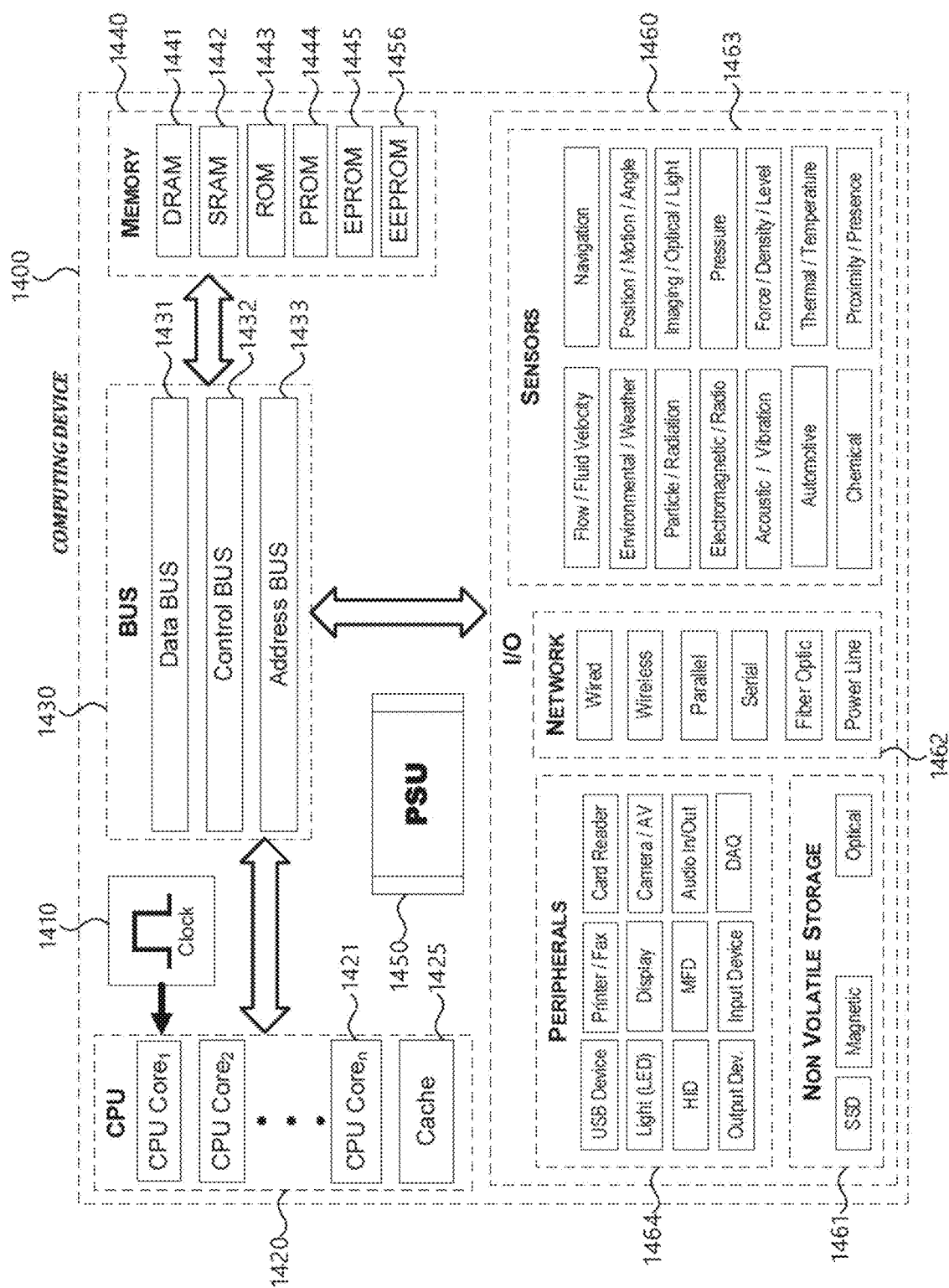


FIG. 14

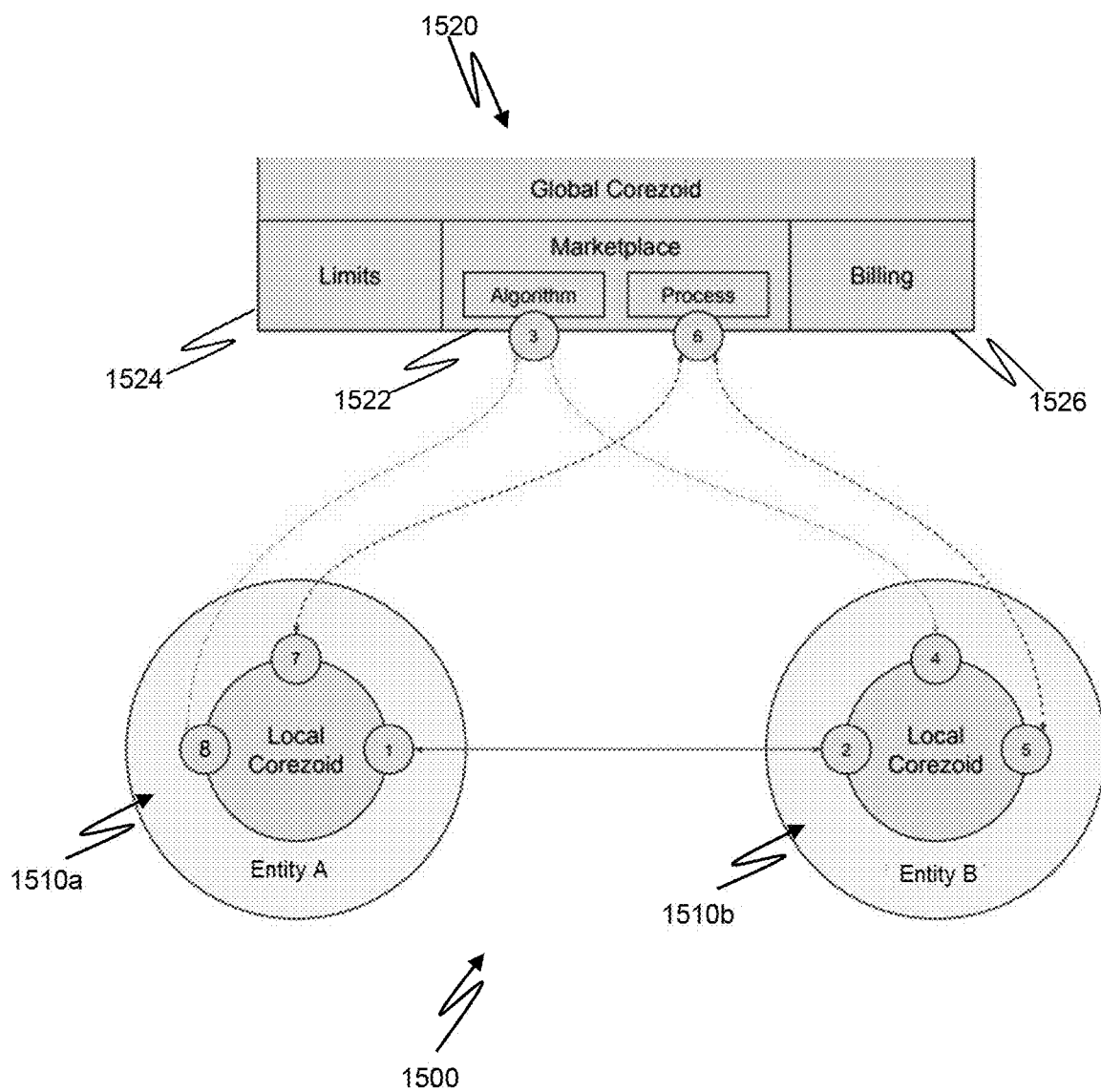


FIG. 15

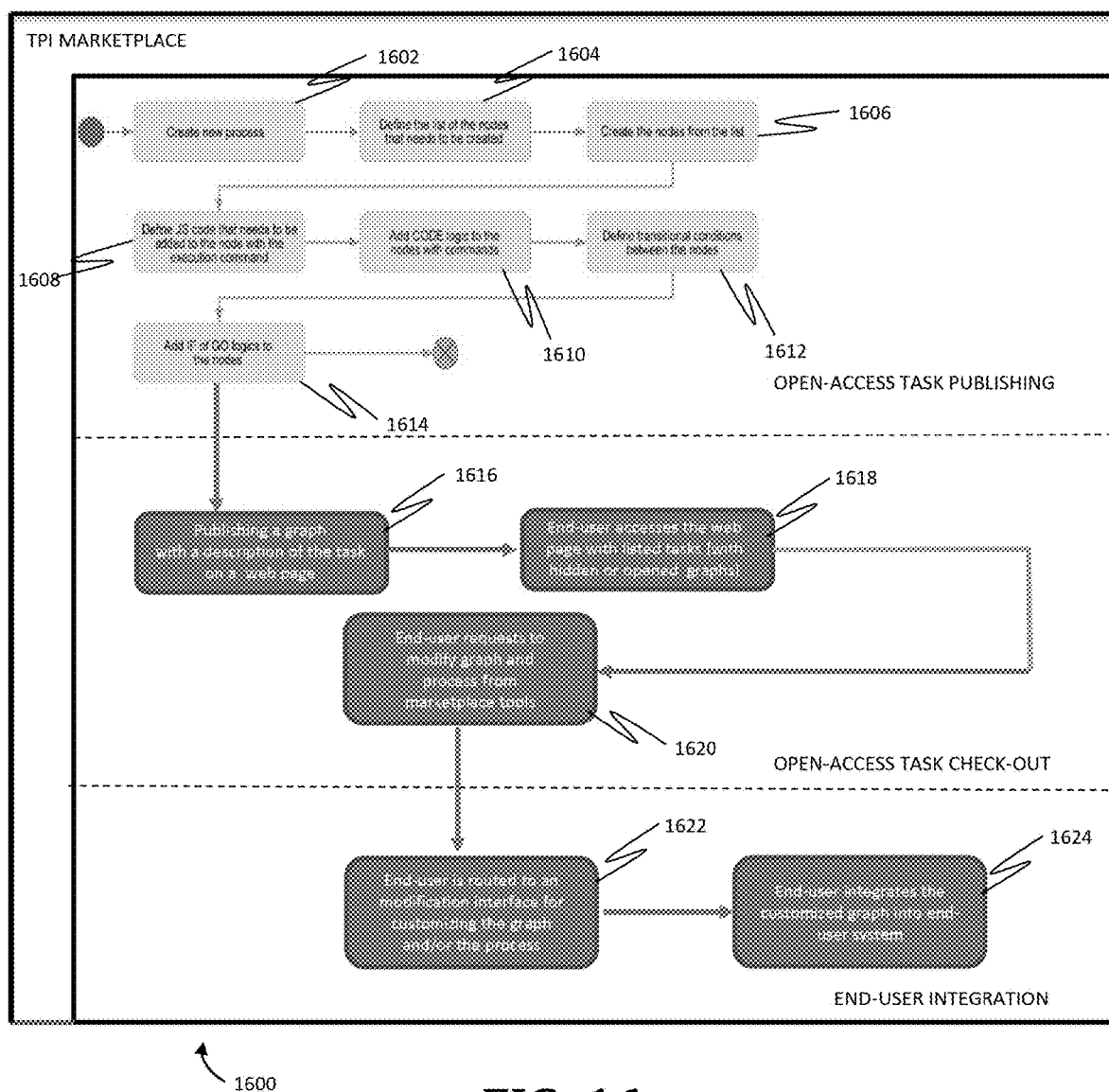


FIG. 16

SYSTEM AND METHOD FOR MODIFYING A DATA PROCESSING METHOD

RELATED APPLICATIONS

[0001] This application is a Continuation-In-Part application under 35 U.S.C. § 120 of U.S. application Ser. No. 17/388,341 filed on Jul. 29, 2021, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/213,114 filed on Jun. 21, 2021.

[0002] Related U.S. application Ser. No. 17/388,341 filed on Jul. 29, 2021 is a Continuation-In-Part application under 35 U.S.C. § 120 of U.S. application Ser. No. 16/252,075 filed on Jan. 18, 2019, which is a Continuation-In-Part application of U.S. application Ser. No. 15/268,802 filed on Sep. 19, 2016, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 62/221,124 filed on Sep. 21, 2015.

[0003] Related U.S. application Ser. No. 16/252,075 filed on Jan. 18, 2019 is also a Continuation-In-Part application under 35 U.S.C. § 120 of U.S. application Ser. No. 15/077,626 filed on Mar. 22, 2016, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 62/137,079 filed on Mar. 23, 2015.

[0004] Related U.S. application Ser. No. 17/388,341 filed on Jul. 29, 2021 is a Continuation-In-Part application under 35 U.S.C. § 120 of U.S. application Ser. No. 16/027,926 filed on Jul. 5, 2018, which is a Continuation-In-Part application of U.S. Ser. No. 15/268,802 filed Sep. 19, 2016, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 62/221,124 filed on Sep. 21, 2015.

[0005] Related of U.S. application Ser. No. 16/027,926 filed on Jul. 5, 2018 is also a Continuation-In-Part application of U.S. application Ser. No. 15/077,626 filed Mar. 22, 2016, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 62/137,079 filed on Mar. 23, 2015.

[0006] Each of the referenced applications is hereby incorporated by reference in its entirety. It is intended that the referenced applications may be applicable to the concepts and embodiments disclosed herein, even if such concepts and embodiments are disclosed in the referenced applications with different limitations and configurations and described using different examples and terminology.

FIELD OF DISCLOSURE

[0007] The present disclosure relates to computer-implemented systems and methods for multi-system connectivity and automation. More specifically, the present disclosure relates to multi-system connectivity and automation via collections of one or more universal computing elements, each of which may comprise an object queue, one or more counters, and a function operating on parameters of objects in the object queue. Furthermore, the invention particularly pertains to altering an already created process flow that uses the one or more universal computing elements by end users to integrate the created process flow into the automation channels of the end users.

BACKGROUND

[0008] In many ways, existing software and business process development paradigms can be rigid and siloed,

burdened by inertia, slow to provide actionable business intelligence, difficult to scale, and difficult for non-coders to interact with.

[0009] Traditionally, many such systems have been developed without much consideration of how to combine them with each other, especially between different vendors (both hardware and software). Many of the configuration values that are parameters of business processes are hardcoded, making changes slow and expensive. The inertia of a heavy, slow-to-change, existing codebase can siphon information technology budgets away from innovation. In addition, many existing systems are batch-based and their response is determined by the batch frequency, not by the fluid, breathing reality of realtime data. Many existing systems also struggle with scalability and concurrency.

[0010] Further, some software development paradigms allow for powerful, flexible algorithm development, and some allow for no-code/low-code approachability. It may be advantageous for a system to provide both. Indeed, systems that may variously address the above issues, and others, may provide advantages over the traditional approaches.

[0011] In view of at least the above shortcomings, a need exists for systems and methods for multi-system connectivity and automation via universal computing elements.

BRIEF OVERVIEW

[0012] This brief overview is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This brief overview is not intended to identify key features or essential features of the claimed subject matter. Nor is this brief overview intended to be used to limit the claimed subject matter's scope.

[0013] Systems and methods for multi-system connectivity and automation via Universal Computing Elements (UCEs) may be provided. A UCE consistent with the embodiments of the present disclosure may be operative with various methods, systems, and computer-readable media (collectively referred to herein as the "platform" or the "disclosed platform").

[0014] The platform may comprise one or more objects, each of which may be characterized by its parameters. The platform may comprise one or more universal computing elements (UCEs), each of which may comprise a queue for objects, one or more counters related to objects in the queue, and a function operating on object parameters and/or counters.

[0015] UCEs may operate in a manner analogous to state machines. Each object in the queue of a UCE may be conceptualized as being in a particular state at any particular time. Such a state may comprise the object's current parameters and their values. Such parameters may include an identifier for the UCE in which the object is queued. A UCE itself may also be conceptualized as being in a particular state at any particular time. Such a state may comprise the object constituents of its queue and the current values of counters related to objects in the queue. A UCE's function may transition objects to new states.

[0016] By way of non-limiting example, a function may change the state of an object by adding, modifying, or removing a parameter of that object. By way of another non-limiting example, a function of a first UCE may change the state of an object so that it may "flow" or "proceed" from the first UCE to a second "interconnected" UCE by chang-

ing a parameter of that object from identifying the first UCE to identifying the second UCE. Both of the above example state changes may be mutually consistent with each other and, indeed, may in some embodiments be engendered by the operation of a single UCE's function.

[0017] UCEs (also sometimes referred to as "nodes") may be connected in myriad ways to form processes, which may be thought of as collections of interconnected UCEs. In some embodiments, the connectivity of UCEs may be simple and linear. In one example, objects may proceed from a starting node through one or more subsequent nodes to an end node, functions operating on objects each step of the way. In further embodiments, the connectivity of UCEs (and of processes), need not be simple nor linear. The interconnections between UCEs (and between processes) may be complex, including loops, branches, nesting, and multiple end points.

[0018] Objects (also sometimes referred to as "tasks"), characterized by parameters (for example, name:value pairs), may flow into a node's queue from a variety of sources. The variety of sources may include, but not be limited to, for example, other processes, application programming interfaces (APIs) and various data sources (databases, etc.), and human input.

[0019] Consistent with embodiments disclosed herein, a node may comprise at least one counter. A node's counters may be used to, by way of non-limited example, keep track of data about the objects currently in that node's queue. A non-limiting example of data related to which an object may keep a counter is the number of objects in the queue. Another such example may be the timestamp of each object's entry into the queue. Another such example may be how many of the objects have parameters that meet particular criteria.

[0020] A function (which, in certain contexts may be disclosed or utilized as at least one "rule"), consistent with the embodiments herein may be associated with a node. The function may be, for example, but is not limited to, a conditional, API call, code snippet, or call to another process. Functions may be configured to operate on (parameters of) objects that enter a UCE's queue or on the UCE's counters. In some embodiments, functions may serve to alter parameters of objects, as may be the case when an API is called and returns pertinent data to add to the object about which it was called.

[0021] The platform may provide, by way of non-limited example, an API interface, analytics, dashboarding, and data visualization functionality. The platform may provide various measures of, and interfaces to, processes in realtime.

[0022] One objective of the disclosed platform may be to provide a UCE comprising a queue of objects, object counters, and a function operating on at least one of the objects.

[0023] Another objective of the disclosed platform may be to provide for a plurality of connected UCEs capable of assembling algorithms of arbitrary complexity.

[0024] Another objective of the disclosed platform may be to provide a Turing complete computational system.

[0025] Another objective of the disclosed platform may be to provide nestable, independent, state machines.

[0026] Another objective of the disclosed platform may be to provide multi-system connectivity and automation of workflows via an interconnected plurality of UCEs.

[0027] Another objective of the disclosed platform may be to connect with a plurality of services and data sources via application programming interface (API).

[0028] Another objective of the disclosed platform may be to provide UCEs having particularized functionalities such as conditionals, API calls, mathematical expressions, variable get/set, and arbitrary code.

[0029] Another objective of the disclosed platform may be to provide a cloud-based platform that can be managed via anonymized/non-identifying data.

[0030] Another objective of the disclosed platform may be to scalably accommodate high request per second workloads.

[0031] Another objective of the disclosed platform may be to provide for no-code/low-code functionality that may enhance the efficiency, speed, or division of labor in software and process development.

[0032] Another objective of the disclosed platform may be to provide rapid prototyping by building complex systems from UCEs.

[0033] Another objective of the disclosed platform may be to provide multi-purpose cloud platform for development, maintenance, management, execution, monitoring and optimization of software for automatic or automated business processes.

[0034] Another objective of the disclosed platform may be to provide process versioning for the platform's processes.

[0035] Another objective of the disclosed platform may be to provide for the possibility of storing data within the platform.

[0036] Another objective of the disclosed platform may be to provide a user interface to the platform that permits "drag-and-drop" creation, selection, interconnection, and modification of UCEs.

[0037] Another objective of the disclosed platform may be to provide a user interface to the platform that provides pertinent information and inputs upon the selection of a UCE.

[0038] Another objective of the disclosed platform may be to provide error handling for API calls and other actions performed by the platform.

[0039] Another objective of the disclosed platform may be to describe changes of the relative states of data within the platform.

[0040] Another objective of the disclosed platform may be to utilize state diagrams in the manner of a database, such that the logic of an object's transition between different states described, and further that state transitions may trigger the launch of processes.

[0041] Another objective of the disclosed platform may be to permit increases in efficiency by shifting a process into an in-memory configuration.

[0042] Another objective of the disclosed platform may be to enable analytics, dashboarding, and data visualization functionality for UCEs.

[0043] Another objective of the disclosed platform may be to enable an end-user to copy an already coded UCE-based task from another user, for example via a marketplace or other portal for sharing already-coded UCE-based tasks.

[0044] Another objective of the disclosed platform may be to enable an end-user to modify of otherwise customize an already coded UCE-based task from another user.

[0045] Another objective of the disclosed platform may be to enable an end-user to integrate an already coded UCE-based task into the task automation channels of the end user.

[0046] Another objective of the disclosed platform may be to provide a Task Publish Integration (TPI) portal for creation of a UCE-based task to be shared with a plurality of other end users.

[0047] Another objective of the disclosed platform may be to provide a TPI portal for publishing of a UCE-based task to be shared with a plurality of other end users.

[0048] Another objective of the disclosed platform may be to provide a Task Publish Integration (TPI) portal for creation of a UCE-based task to be shared with a plurality of other end users.

[0049] Another objective of the disclosed platform may be to provide a TPI portal for copying a UCE-based task that has been shared with a plurality of other end users.

[0050] Another objective of the disclosed platform may be to provide a TPI portal for modifying a UCE-based task to be shared with a plurality of other end users.

[0051] The disclosed platform may provide technical advantages over conventional solutions. One such advantage may be the scalability of UCE-based processes to possibly handle thousands, tens of thousands, or more of concurrent tasks.

[0052] Another such advantage may be the breadth of the platform's trackable data and speed of access of analytics, dashboarding, and data visualization to the same.

[0053] Another such advantage may be the ability for a queue to store an arbitrary number of objects.

[0054] Another such advantage may be the breadth of functionality available to UCE functions.

[0055] Another such advantage may be the possibility for simultaneous, multifarious sourcing of objects.

[0056] Another such advantage may be the possibility of accepting and handling objects with heterogeneous data structures.

[0057] Another such advantage may be the limitless capacity for sequentializing objects in queues via custom counters.

[0058] Both the foregoing brief overview and the following detailed description provide examples and are explanatory only. Accordingly, the foregoing brief overview and the following detailed description should not be considered to be restrictive. Further, features or variations may be provided in addition to those set forth herein. For example, embodiments may be directed to various feature combinations and sub-combinations described in the detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0059] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various embodiments of the present disclosure. The drawings contain representations of various trademarks and copyrights owned by the Applicants. In addition, the drawings may contain other marks owned by third parties and are being used for illustrative purposes only. All rights to various trademarks and copyrights represented herein, except those belonging to their respective owners, are vested in and the property of the Applicants. The Applicants retain and reserve all rights in their trademarks and copyrights included herein, and grant permission to reproduce the material only in connection with reproduction of the granted patent and for no other purpose.

[0060] Furthermore, the drawings and their brief descriptions below may contain text or captions that may explain

certain embodiments of the present disclosure. This text is included for illustrative, non-limiting, explanatory purposes of certain embodiments detailed in the present disclosure. In the drawings:

[0061] FIG. 1A illustrates a conceptual diagram of a universal computing element in accordance with various embodiments of the present disclosure.

[0062] FIG. 1B illustrates a conceptual diagram of a process comprising universal computing elements in accordance with various embodiments of the present disclosure.

[0063] FIG. 1C illustrates a conceptual diagram of a process comprising universal computing elements in accordance with various embodiments of the present disclosure.

[0064] FIG. 2A illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0065] FIG. 2B illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0066] FIG. 3 illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0067] FIG. 4 illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0068] FIG. 5 illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0069] FIG. 6 illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0070] FIG. 7 illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0071] FIG. 8A illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0072] FIG. 8B illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0073] FIG. 8C illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0074] FIG. 9A illustrates a flowchart diagram of an example method in accordance with various embodiments of the present disclosure.

[0075] FIG. 9B illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0076] FIG. 9C illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0077] FIG. 10 illustrates a block diagram of the architecture of an example implementation of the disclosed platform in accordance with various embodiments of the present disclosure.

[0078] FIG. 11 illustrates a block diagram of the architecture of an example implementation of the disclosed platform in accordance with various embodiments of the present disclosure.

[0079] FIG. 12 illustrates simplified activity diagram of the interpretation algorithm in an example implementation of the disclosed platform in accordance with various embodiments of the present disclosure.

[0080] FIG. 13A illustrates a block diagram of an example implementation of the disclosed platform in accordance with various embodiments of the present disclosure.

[0081] FIG. 13B illustrates an example interface to the disclosed platform showing an example implementation in accordance with various embodiments of the present disclosure.

[0082] FIG. 14 illustrates a block diagram of a platform including a computing device in accordance with various embodiments of the present disclosure.

[0083] FIG. 15 illustrates an operating environment illustrating the extent of interaction between companies employing the node-based finite-state task loop.

[0084] FIG. 16 illustrates a process flow of a task publish-integration (TPI) marketplace in accordance with an aspect of the invention.

DETAILED DESCRIPTION

[0085] As a preliminary matter, it will readily be understood by one having ordinary skill in the relevant art that the present disclosure has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the disclosure and may further incorporate only one or a plurality of the above-disclosed features. Furthermore, any embodiment discussed and identified as being “preferred” is considered to be part of a best mode contemplated for carrying out the embodiments of the present disclosure. Other embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present disclosure.

[0086] Accordingly, while embodiments are described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present disclosure and are made merely for the purposes of providing a full and enabling disclosure. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection be defined by reading into any claim a limitation found herein that does not explicitly appear in the claim itself.

[0087] Thus, for example, any sequence(s) and/or temporal order of stages of various processes or methods that are described herein are illustrative and not restrictive. Accordingly, it should be understood that, although stages of various processes or methods may be shown and described

as being in a sequence or temporal order, the stages of any such processes or methods are not limited to being carried out in any particular sequence or order, absent an indication otherwise. Indeed, the stages in such processes or methods generally may be carried out in various different sequences and orders while still falling within the scope of the present disclosure. Accordingly, it is intended that the scope of patent protection is to be defined by the issued claim(s) rather than the description set forth herein.

[0088] Embodiments of the present disclosure may provide a hardware and software platform operative by a set of methods and computer-readable media comprising instructions configured to operate computing elements in accordance with the methods. The present disclosure depicts and/or describes at least one example method of a plurality of methods that may be performed by at least one of the aforementioned computing elements. Various hardware components may be used at the various stages of operations disclosed with reference to each computing element.

[0089] For example, although methods may be described to be performed by a single computing device, it should be understood that, in some embodiments, different operations may be performed by different networked elements in operative communication with the computing device. For example, at least one computing device may be employed in the performance of some or all of the stages disclosed with regard to the methods. Similarly, an apparatus may be employed in the performance of some or all of the stages of the methods. As such, the apparatus may comprise at least those architectural components as found in the computing device.

[0090] Furthermore, although the stages of the example method(s) disclosed herein are disclosed in a particular order, it should be understood that the order is disclosed for illustrative purposes only. Stages may be combined, separated, reordered, and various intermediary stages may exist. Accordingly, it should be understood that the various stages, in various embodiments, may be performed in arrangements that differ from the ones claimed below. Moreover, various stages may be added or removed from the without altering or deterring from the fundamental scope of the depicted methods and systems disclosed herein.

[0091] Additionally, it is important to note that each term used herein refers to that which an ordinary artisan would understand such term to mean based on the contextual use of such term herein. To the extent that the meaning of a term used herein—as understood by the ordinary artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by the ordinary artisan should prevail.

[0092] Regarding applicability of 35 U.S.C. § 112, ¶6, no claim element is intended to be read in accordance with this statutory provision unless the explicit phrase “means for” or “stage for” is actually used in such claim element, whereupon this statutory provision is intended to apply in the interpretation of such claim element.

[0093] Furthermore, it is important to note that, as used herein, “a” and “an” each generally denotes “at least one,” but does not exclude a plurality unless the contextual use dictates otherwise. When used herein to join a list of items, “or” denotes “at least one of the items,” but does not exclude

a plurality of items of the list. Finally, when used herein to join a list of items, “and” denotes “all of the items of the list.”

[0094] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While many embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods. Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of the disclosure is defined by the appended claims. The present disclosure contains headers. It should be understood that these headers are used as references and are not to be construed as limiting upon the subjected matter disclosed under the header.

[0095] The present disclosure includes many aspects and features. Moreover, while many aspects and features relate to, and are described in, the context such as authentication workflows, communication workflows, chat and chatbot workflows, finance and banking workflows, and logistics workflows, embodiments of the present disclosure are not limited to use only in this context. Other examples may include customer relationship management workflows, ecommerce workflows, robotics workflows, rideshare and transportation workflows, software development workflows, etc.

I. Overview

[0096] Consistent with embodiments of the present disclosure, systems, methods, and computer-readable media for multi-system connectivity and automation via universal computing elements—platform 100—are provided. Various embodiments of platform 100 are described herein. Components of platform 100 as presented in the following disclosure may be integrated, used independently, in conjunction with, used separately, or in connection with other embodiments they are not shown or described as functioning with. Any aspects of one embodiment may or may not be used interchangeably with other elements and aspects of a platform 100 as presented in the present disclosure.

[0097] Some or all of the following components may be present in embodiments of the present disclosure. The below description is in no way intended to limit the components that may be present in addition or in alternative to the listed components, nor to require that any particular component be included in a form described below.

[0098] a. Platform 100

[0099] Platform 100 may, in various aspects and embodiments, be conceptualized as a collection of one or more interconnected universal computing elements (or “UCEs”) 110. An example conceptual diagram of a UCE (or “node”) 110 is depicted in FIG. 1A. Consistent with the depiction in FIG. 1A, UCEs 110 may comprise a queue 120 for objects (or “tasks”) 130. Objects 130 may be characterized by parameters 140 (e.g., name:value pairs). Counters 150 may store data about queue 120 or objects 130. A function (or “rule”) 160 may operate on parameters 140 of the UCE’s 110 objects 130 or on counters 150 of the UCE 110.

[0100] UCEs 110 may operate in a manner analogous to state machines. Each object 130 in the queue 120 of a UCE 110 may be conceptualized as being in a particular state at any particular time. Such a state may comprise the object’s 130 current parameters 140 and their values. Such parameters 140 may include an identifier for the UCE 110 in which the object is queued. A UCE 110 itself may also be conceptualized as being in a particular state at any particular time. Such a state may comprise the object 130 constituents of its queue 120 and the current values of counters 150 related to objects 130 in the queue 120. A UCE’s 110 function 160 may transition objects 130 to new states. By way of non-limiting example, a function 160 may change the state of an object 130 by adding, modifying, or removing a parameter 140 of that object 130.

[0101] In various embodiments, a UCE 110 may form an outbound connection with another UCE 110 or with itself. By way of another non-limiting example, a function 160 of a first UCE 110 may change the state of an object 130 so that it may “flow”, “proceed”, or “be released” from the first UCE 110 to a second “interconnected” UCE 110 by changing a parameter 140 of that object 130 from identifying the first UCE 110 to identifying the second UCE 110. The second UCE 110 may be said to “accept” the object 130. By way of another non-limiting example, an object 130 may remain in a UCE’s 110 queue 120 until acted upon by the UCE’s 110 function 160. By way of another non-limiting example, an object 130 may remain in the queue 120 an “end” node 110 indefinitely.

[0102] UCEs 110 may be combined together into collections of UCEs 110 that form a process 170. In a process 170, objects 130 may be routed among UCEs 110, passing through their respective queues 120, wherein their parameters 140 may be operated upon and modified by the UCEs’ 110 respective functions 160. The path may be simple, as depicted for example in FIG. 1B, but need not be. Processes 170 may be arbitrarily complex, including include loops, branches, nesting, multiple endpoints, etc., as depicted for example in FIG. 1C (which also depicts calls, represented by dotted lines, from UCEs 110 to other processes 170).

[0103] FIGS. 2A-B depict an example interface to platform 100 showing example implementation 200. In this example implementation 200, UCE nodes 110 (labeled for clarity in FIG. 2B with an “N”) are collected into an example process 170 for authenticating a user via their phone number. This example implementation 200 depicts nodes 110 sub-specialized (as discussed further infra) and variously connected. Also depicted are various nodes’ 110 pertinent parameters 140 (see boxes called out by arrows). This example implementation 200 is explored in greater detail infra following a brief overview of some features platform 100 may comprise, as well as some examples involving the same.

[0104] i. Brief Overview of Features Platform 100 May Comprise

[0105] By way of non-limiting example, a queue 120 may be generally conceptualized, in various embodiments, as representing a collection of tasks 130 to or about which a node 110 is to do something (i.e., via function 160). Objects 130 may enter a UCE’s 110 queue 120 from a variety of sources, which may include the results of an API call 131 (which may be generalized as encompassing a broad array of data sources and external processes), human input 132, another UCE 110a, or another process 170a.

[0106] By way of non-limiting example, objects **130** may be generally conceptualized, in various embodiments, as representing a discrete conceptual unit about which something is to be done. Objects **130** may be generally conceptualized this way both in the sense of an individual UCE **110** “doing” something, in its course of operation, to or about object **130**, and in the more general sense of object **130** existing in process **170** as a task **130** to be purposefully acted upon. Some examples may include as a caller to be routed to an operator, a loan borrower to be checked for creditworthiness, a chat participant to be authenticated, etc. More abstract examples may include a currency exchange rate to be polled, location-specific weather forecast to be queried, an HTTP connection to be kept alive, etc.

[0107] By way of non-limiting example, parameters **140** may be generally conceptualized, in various embodiments, as characterizing the current state of an object **130**, such as, for example, a caller’s initial call timestamp, a loan borrower’s tax ID, a chat participant’s chat instance ID, an exchange rate request’s from and to currencies, a weather forecast request’s geographic information, a Hypertext Transfer Protocol (HTTP) connection’s Uniform Resource Locator (URL), etc.

[0108] Still consistent with embodiments of the present disclosure, parameters **140** may be added or modified by UCEs **110**. For example while a caller’s initial call timestamp may stay the same throughout a process **170**, a parameter **140** signifying the number of times the caller has been transferred may increment. An object **130** representing a chat participant may be updated with the parameter **140** of the participant’s response to a request for their phone number to authenticate them. An object **130** representing a weather forecast request may be updated with the parameter **140** of current temperature, returned from a call to a weather API. In another non-limiting example, as an object **130** proceeds through a process **170**, parameters **140** may be altered by a first UCE **110**. If the object **130** passes into the queue **120** of a subsequent UCE **110**, the subsequent UCE **110** may “see” the parameters **140** of the object **130** as altered by the first UCE **110**.

[0109] By way of non-limiting example, counters **150** may be generally conceptualized, in various embodiments, as counting or ordering objects **130** with respect to queue **120**, such as the number of objects **130** in queue **120**, the timestamp of entry of an object **130** into the queue **120**.

[0110] By way of non-limiting example, a function **160** may be generally conceptualized, in various embodiments, as the aspect of a UCE **110** that is doing the “doing” that UCE **110** does to or in association with objects **130** that enter its queue **120**. Functions **160** may comprise operations such as mathematical expressions, conditionals, getting and setting variables, running code (e.g., Javascript, Java, Erlang, etc.).

[0111] In yet further embodiments, functions **160** may be configured for establishing connections to external systems (e.g., API calls, database connections, code versioning repositories), calling or modifying other processes **170**, asking for input from a user, etc. A function **160** may, as part of its operation, alter one or more parameters **140** of an object **130** to which it is applied.

[0112] In various embodiments, upon the execution of a first UCE’s **110** function **160**, an object **130** may pass to a subsequent one of the plurality of UCEs **110b** (which may include a loopback to the same UCE **110**). Subsequent UCE

110b may apply a function **160** to the object **130**. Additionally or alternatively, subsequent UCE **110b** may be an end state (i.e., an end node **110**). In various embodiments, there may be at least one interconnection between each UCE **110** and at least one other UCE **110**, each interconnection comprising an outbound UCE **110** and an interconnection function of the outbound UCE **110** identifying one of the plurality of UCEs **110b**.

[0113] By way of non-limiting example, a first UCE’s **110** function **160** may comprise an interconnection that may identify one of the plurality of UCEs **110b** that may comprise a process **170**. The interconnection may be configured to set one of the one or more parameters **140** of a queued object **130** to which the function **160** is applied to the UCE **110b** identified by the interconnection. In this example, setting a “node identifier” parameter **140** to UCE **110b** may precipitate the release of the object **130** from the queue **120** of the first UCE **110** to the queue **120** of the subsequent, identified UCE **110b**.

[0114] In some embodiments, an object **130** may pass to one of potentially many UCEs **110b**, depending on the one to arbitrarily many exit states possible for a function **160**. This can be as simple as shunting off error-state objects **130** to an error handling node **110**, or as complex as logic/code dictates.

[0115] For example, function **160** may set a parameter **140** of objects **130** that pass through its queue **120**. As another example, function **160** may get an object **130** from the queue **120** of a UCE **110** from another process **170**. As another example, function **160** may apply a conditional to a parameter **140** of each object **130** that enters the queue **120**, such as whether its value exceeds a threshold, or whether the returned text string matches one of various string values. The outcome of a conditional function **160** may be used to direct the flow of outgoing objects **130** to interconnected nodes **110**. As an example, a conditional may have a plurality of outcomes, each outcome providing an outbound interconnection to a node **110**.

[0116] In a non-limiting, general example of the operation of a UCE **110**, queue **120** may take in objects **130** representing prospective loan borrowers, via manual data entry **132**, the objects **130** having parameters **140** such as name and tax ID. UCE **110** may apply the function **160** of performing an API call **161** to a credit reporting service, which may return data concerning each object’s **130** (borrower’s) creditworthiness. An implementation of platform **100** consistent with this example and comprising a user interface may enable a user to configure the API call UCE **110** in a manner such as that depicted in FIG. 4.

[0117] In a non-limiting, general example of the operation a process **170**, a first UCE **110** may take in objects **130**. The objects’ **130** parameters **140** may comprise user data and message text. The first UCE **110** may pass objects **130** to a second UCE **110** that may query a database for a phone number corresponding to the objects **130** (users). The second UCE may pass objects **130** (now including phone number as a parameter **140**) to a third UCE **110** that may call an API for an SMS messaging service, instructing the SMS messaging service to transmit the textual data to the user’s phone number. An implementation of platform **100** consistent with this example and comprising a user interface may enable a user to configure these UCE **110** in a manner such as that depicted in FIGS. 2A-B, 4, and 7.

[0118] In this example, there may be UCE 110 branches performing various functionality—for instance, handling error states of the various steps along the path. The second UCE 110 may have an error handling branch to a node 110 that handles, for example, what happens if the database query takes too long to respond, finds the user but no phone number, or fails to find a corresponding user at all. The third UCE 110 may have an error handling branch to a node 110 that handles, for example, what happens if the API takes too long to respond, or returns an error due to an out-of-country or incorrectly formatted number.

[0119] Platform 100 may comprise a user interface for creating, managing, and interacting with UCEs 110 and processes 170. Some features of a user interface, as may be present in various embodiments, may follow. One such feature may be foldered organization and management of processes 170. Another such feature may be “drag-and-drop” (i.e., pointing device navigable) creation, selection (as among nodes 110 differentiated by function 160), interconnection, and modification of UCEs 110. Another such feature may be the display of process or state “maps” that visualize relationships between nodes 110. In various embodiments, a visual mapping of a process 170 may comprise displaying each of the process’ 170 UCEs 110 and each interconnection among the process’ 170 UCEs 110.

[0120] Another such feature may be providing templates that comprise pre-set and preconnected collections of UCEs 110. In various embodiments, such templates may be presented to a user of an example interface to platform 100 in the form of a template library. Another such feature may be providing model API call nodes 110 preconfigured for connecting with particular APIs. Another such feature may be the display of pertinent data and inputs upon indicating (i.e., selecting) a particular element in a process 170 such as a node 110 or connection.

[0121] In an example, a user interface of platform 100 might display, upon selection of a particular node 110, pertinent parameters 140, fields for entry of pertinent data (e.g., a language selector and code or database query input box, URL for an API call). The user interface may also display other pertinent options for the functioning of the node 110, such as whether to send system information, whether and what to send with respect to header information, cryptographic signature, limiting the time of a task 130 in the node 110, etc.

[0122] In some embodiments, a user interface of platform 100 may automatically create and interconnect various nodes 110 upon the user’s creation of a node 110 of a particular type. For example, upon creating a UCE 110 of the type “API call”, platform 100 may automatically create an error handling loop comprising a conditional node 110 which may operate on the type of error (e.g., long wait for answer, failed to add parameters 140 to query, invalid format provided, empty URL, response is too large, etc.). The conditional node 110 may, by its logic, pass objects 130 with some errors to a delay node 110 (looped back to the API call node 110 to retry after a delay). This may be the case for errors that are deemed “recoverable” (e.g., long wait for answer). The conditional node 110 may, by its logic, pass other errors to a terminal exit node 110 signifying the process 170 was terminated in an error state. This may be the case for errors that are deemed “non-recoverable” (e.g., empty URL). An implementation of platform 100 consistent with this example and comprising a user interface may

enable a user to configure an API call UCE 110 and a conditional UCE 110 in a manner such as that depicted in FIGS. 4 and 6.

[0123] Platform 100 may comprise analytics, dashboarding, and data visualization functionality. In various embodiments, platform 100 may provide visibility into the current status of running processes 170. For example, platform 100 may provide metrics that may be derived from counters 150 or object 130 parameters 140. Examples of metrics might include how many callers are currently in a call queue waiting for an operator to speak with them, how many loan requests have been routed to various risk tier “buckets” (e.g., end nodes 110 to which objects 130 are routed), or the 24 hour trend of currency exchange rates polled at regular intervals.

[0124] In various embodiments of platform 100, counters 150 as well as the states (current parameters 140) of objects 130, of any UCE 110, may be utilized for current and historical analytics, dashboards, data visualizations, graphical representations, etc. Such status visibility may, in some embodiments, be available in “realtime”. In some embodiments, “realtime” may mean there is little delay between platform 100 reaching a state and the availability of that current state to analytics/dashboarding.

[0125] Platform 100 may itself comprise one or more APIs for interacting with various components of platform 100. Non-limiting examples of API interactions with platform 100 follow. An example API to platform 100 may enable the creating, duplicating, and destroying of objects 130 and the moving objects 130 between nodes 110. An example API to platform 100 may enable altering parameters 140. An example API to platform 100 may enable altering counters 150. An example API to platform 100 may enable creating, modifying, and deleting processes 170 (including various aspects of processes’ 170 UCEs 110). An example API to platform 100 may enable creating, modifying, deleting, and interacting with analytics, dashboards, and data visualizations. An example API to platform 100 may enable the performing of administrative functions (e.g., changing user permissions).

[0126] Platform 100 and elements thereof (e.g., a process 170), may be implemented in a storage medium such as a database, or in memory, as is discussed further infra Section III (see FIGS. 10-11). UCEs 110 and processes 170 may be combined to construct algorithms of arbitrary complexity; in various embodiments, platform 100 may be implemented as a Turing complete computational system, as is discussed further infra Section III (see FIGS. 13A-B).

[0127] ii. Example Implementation 200

[0128] An example implementation 200 consistent with that depicted in FIGS. 2A-B includes a plurality of nodes 110 interconnected to form a process 170 for authenticating users via their phone numbers. This example may begin with a “Start” node 110 where objects 130 (representing users to be authenticated) may enter process 170. An object 130 may proceed to a “Copy task” node 110 (labeled “1”), where it may be copied, along with relevant parameters 140, to another process 170. In this example, the object 130 may be copied to a “Send Message” process 170 for sending an authentication SMS message (e.g., a 6-digit code) to the user, so that they may confirm their receipt of the message and thus their possession of the relevant authenticating device.

[0129] If the communication with the “Send Message” process 170 results in an error, an object 130 may proceed into a special error handling loop that may delay and retry or exit altogether in an error state. This may reflect a specific instance of how platform 100 may, in some embodiments, enable error handling, in that nodes 110 whose functions 160 that may result in error states may be provided with “conditional-like” interconnection points that may permit outflow of objects 130 based on the occurrence of an error state.

[0130] Otherwise, objects 130 may proceed to a “Waiting for Callback” node 110 (labeled “2”) that awaits the response of the user to the authentication message generated by the “Send Message” process 170. Any user response may proceed to the “Condition” node 110 (labeled “3”) that may provide an opportunity for an exit command to be issued by the user. Such input may lead to a terminal “exit” node 110, canceling the process 170. Otherwise, an object 130 may proceed to another “Condition” node 110 (labeled “4”) which may validate the user’s response. If the user entered the wrong value, object 130 may return to the “Copy task” node 110 (labeled “1”). Otherwise, upon success, object 130 may proceed to one of two “Set Parameter” nodes 110 (labeled “5.1” and “5.2”) depending how the platform that served as the source of the object 130 provided phone contact details. If for some reason the setting of the “phone” parameter 140 would result in an error state, each of the “Set Parameter” nodes 110 has a terminal exit node 110.

[0131] An object 130 may then proceed to a “Modify Task” node 110 (labeled “6”). This node 110 may set the “phone” parameter 140 of a user data object 130 in a “User Profile” process 170. There may be an error handling loop to deal with the possibility that communication with the “User Profile” process 170 may result in an error. An object 130 may then proceed to a “Copy task” node 110 (labeled “7”). This node 110 may copy the object 130 (user), now authenticated, to another process 170. In this example, this may be a “Router” process 170 from which this authentication process 170 may have been initially called. Again, there may be an error handling loop attached to this node 110 (for example, if the other process 170 is not active) but if no error is encountered, an object 130 may then proceed to a successful “END” node 110.

[0132] It may be noted that, in various embodiments, each of the nodes 110 in a process 170 such as that depicted in example implementation 200 may be trackable in a dashboard or otherwise available as a source of analytics data. For example, a dashboard may track a counter 150 of how many users issued an exit command before reaching a successful conclusion (objects 130 that reach the exit node 110 of the “Condition” node 110 labeled “3”), how many users there are currently waiting for a response to the “Send Message” process 170 (objects 130 in the queue 120 of the “Waiting for Callback” node 110 labeled “2”), how many users successfully authenticated (objects 130 that reach the “END” exit node 110 of the “Copy task” node 110 labeled “7”), etc.

[0133] Further discussion of features platform 100 may comprise follows.

[0134] b. Universal Computing Element 110 and Process 170

[0135] Platform 100 may comprise a universal computing element 110. A UCE 110 may comprise a queue 120 of objects 130 characterized by parameters 140, counters 150

that may store data about queue 120 or objects 130, and a function 160 that may operate on parameters 140 or counters 150.

[0136] In some embodiments, UCEs 110, as presented to users in an interface to platform 100, may be specialized so as to, for example, simplify and guide the user experience. Such specialization may, for instance, comprise specialized function 160 functionality that presents particular “types” of nodes 110 (discussed further infra) such as conditional nodes 110, API call nodes 110, delay nodes 110, wait for response nodes 110, etc., each of which may be a subset of the generalized power of function 160 to operate upon parameters 140 and counters 150.

[0137] Inputs of objects 130 to a process 170 (and generally to a UCE 110) may come from multiple sources. For example, 4 different APIs 131 and a manual data entry source 132 may all feed the start node 110 of a single process 170. In various embodiments, one or multiple processes 170 may have a node 110 whose function 170b calls on a (sub-) process 170 to perform some subset of processing. As an example, such a “calling” process 170 may act as a source 170a of (i.e., “passing in” of) objects 130 to the (sub-) process 170.

[0138] UCEs 110 may have between zero (i.e., an end node 110) and arbitrarily many outbound paths to other UCEs 110. Multiple UCEs 110 may have outbound interconnections to a single inbound (i.e., receiving) UCE 110. UCEs 110 can be connected linearly or in complex looping, branching, etc. configurations. Platform 100 may comprise protections against infinite loops, for example a rule that stops processing if a task 130 would be cycled over 50 times through a loop of nodes 110.

[0139] In various embodiments, one or more end nodes 110 may serve as the terminal points of a process 170. By way of non-limiting example, an end node 110 may have zero outbound interconnections to other UCEs 110. Consistent with such an example, objects 130 that reach (the queue 120 of) an end node 110 may not proceed to any other nodes 110, but rather may be “collected” in the end node’s 110 queue 120. In some embodiments not inconsistent with such an example, an object 130 reaching an end node 110 may trigger an action by the platform 100, such as issuing an alert, creating a new object 130, or invoking another process 170.

[0140] In some embodiments, one or more counters 150 may track objects 130 that reach an end node 110. In an example, an end node 110 may have a counter 150 that tracks the number of objects 130 in that end node’s 110 queue 120. In another example, an end node 110 may have a counter 150 that tracks the time of entry of each object 130 to that end node’s 110 queue 120.

[0141] Processes 170 may have the ability (e.g., through a user interface selection) to be in different states of activity. Different states of activity may comprise, but not be limited to, an active state (i.e., actively running and processing objects 130), a paused (i.e., not running) state, or debug state (which may permit additional development activities in the running of processes 170 such as setting breakpoints). A process 170 in a paused state may cause an error state in another process 170 that calls it or otherwise depends on it being in active operation. This may make it opportune for some embodiments to comprise a user interface to platform 100 that automatically creates error handling loops for functions 160 like calling another process 170.

[0142] c. Queue 120, Object 130, and Parameters 140

[0143] Platform 100 may comprise a queue 120 of one or more objects 130. An object 130 may comprise data in the form of parameters 140 that characterize object 130. Object 130 may be devoid of parameters 140 (i.e., a pure “pulse” to be run through the process 170). A queue 120 may store an arbitrary number of objects 130.

[0144] While objects 130 in a queue 120 may correspond to ordering information (e.g., a timestamp counter 150 sequentializing the entry of objects 130 into the queue 120), a queue 120 need not be conceptualized as a sequential first in first out (FIFO) or last in first out (LIFO) data structure. Objects 130 are susceptible to infinitely many categorizing criteria based on their parameters 140 and various counters 150. This categorizing principle may be used to distribute objects 130 amongst UCEs 110 (recall that an object can pass to one of potentially many UCEs 110b) in ways that do not depend on the time sequence of entry of the objects 130 into the queue 120.

[0145] In some embodiments, some parameters 140 may be “global” in the sense that these parameters 140 are available as to all objects 130 at all times. Such globally available parameters 140 may include a timestamp of the object’s 130 creation (i.e., when it first became an object in this process 170), the last time the object 130 changed state (i.e., change in a parameter 140), various static and dynamic timer values, the ID of the current node 110 where the object 130 is queued, and the ID of the object 130 itself.

[0146] Parameters 140 may comprise any type of data, from strings, numbers, and booleans, to geographic coordinates, JavaScript Object Notation (JSON) data, images, binary data, pointers, etc. Not all objects 130 in a process 170 or in a queue 120 are required to have the same data structure (i.e., set of parameters 140 and data types for their values). Since inputs of objects 130 to a process 170 or UCE 110 may come from different sources, various sources may name, type, structure, include, or omit parameters 140 in their own idiosyncratic ways. For example, while one bank’s API may return a field (parameter 140) titled “blacklisted” with a value of “TRUE”, another’s may return the field “black_listed” with a value of “yes”, and another’s may not return the field at all.

[0147] In some senses, “events” can be conceptualized as drivers behind changes in state of objects 130. One example of such driver may include a first credit card transaction acting as a trigger for processes 170 to change a parameter 140 of an object 130 representing the credit card holder from a state (parameter 140 value) of “inactive” to “active”. In keeping with this conceptualization, the absence of an “event” can itself be an event. By way of some non-limiting examples: a credit card holder going 60 days without a credit card transaction, or an online shopper going 24 hours with items in a checkout “basket” without clicking the “check out” button, may be the “event” upon which platform 100 takes action.

[0148] d. Counter 150

[0149] Platform 100 may comprise a counter 150. Counters 150 may store information about queue 120 or object 130. For example, a counter 150 may store information about the number of objects 130 in queue 120. For another example, a counter 150 may store the timestamp of each object’s 130 entrance into queue 120. Counter 150 may store other information (i.e., “custom counters”) about objects 130 in the queue 120. By way of some non-limiting

examples of counters 150: how many objects 130 representing callers have a parameter 140 of “VIP” set to “TRUE”, how many objects 130 representing potential loan customers do not contain a parameter 140 of “credit score”, or how many objects 130 representing internet of things (IOT) devices fall within a particular geographic region.

[0150] Counters 150 may be utilized by processes 170 to set limits according to business logic. For example, a counter 150 of the queue 120 of active chats with a customer service agent may be utilized to triage away additional incoming chat instances (objects 130) once the queue 120 reaches 10. This could service the business logic “agents are allowed to chat with a maximum of 10 customers at one time.” For another example, a counter 150 of the timestamps of entry of tasks 130 into the queue 120 of a node 110 signifying a caller ready to be connected with an operator may be utilized to calculate the time since entry. This could be utilized in implementing the business logic “no caller (who reaches this point in the call system) should wait more than 30 seconds to speak with a human.”

[0151] e. Function 160

[0152] Platform 100 may comprise a function 160. A node’s 110 function 160 may operate on a counter 150 or on a parameter 140 of an object 130 in the queue 120 of that node 110. Functions 160 may be categorized in various ways; examples of functions 160 include calls to APIs 161 and to external (or internal) systems and interfaces in general, calls for manual input 162, calls to other processes 170b (which may include a process 170 calling itself), execution of code 163, calls to databases 164, system actions 165. System actions 165 may comprise such actions as applying a conditional (e.g., comparison operators, logical operators, bitwise operators, regular expressions, etc.), waiting for a response, delaying further processing, error handling, setting a parameter 140, establishing a queue 120 for another process 170 to process, performing mathematical operations, employing a source of randomness (e.g., a hardware or software random number generator), setting a state, and issuing a notification. General UCE 110 functions 160 may comprise counter functions 166 that operate on counters 150. Counter functions 166 may be based upon such counter 150 values as the number of objects 130 in queue 120, the timestamp of objects’ 130 entry into queue 120, or other counters 150. There may be overlap between various categorizations, especially with the relatively encompassing function 160 of executing code 163.

[0153] Some depictions of example interfaces to platform 100, showing example implementations of various functions 160, follow. FIG. 3 depicts an example interface to platform 100 showing example implementation 300. In example implementation 300, an interface 310 may present a “menu” of available types of node 110. In an example user interface to platform 100 consistent with that depicted in FIG. 3, a user may, upon interacting with (e.g., mousing over, clicking) an interconnection between a first and a second node 110, be presented with an interface for creating a new node 110 interposed between the extant interconnected nodes 110. Such an interface may present the user with a “menu” of available types of nodes 110 (i.e., UCEs 110 classified by the type of predetermined function 160 with which they are to be created). Upon the user’s selection of a particular type of node 110 from the “menu”, platform 100 may create a new node 110 interposed between the two extant nodes 110—that

is, the new node 110 may be a recipient of an interconnection from the first node and the originator of an interconnection to the second node 110.

[0154] FIG. 4 depicts an example interface to platform 100 showing example implementation 400. In example implementation 400, an API call node 410 is shown. A node 110 whose function 160 is an API call may be configured to communicate with an external system (which term may include an application programming interface to platform 100 itself) in a structured way. This structure may be reflected in data and inputs presented to a user upon selecting in a user interface to platform 100 of an API call node 110, such as that depicted in FIG. 4. An API call may utilize a URL so that it may communicate with the external system, URL string parameters that may be included in the URL request string, an HTTP request method, etc. The various inputs to this example API call may be presented to a user in a user interface of platform 100.

[0155] In an example consistent with example implementation 400 as depicted in FIG. 4, a user may be provided appropriate inputs for the API call. Such inputs may include the API URL (e.g., a text box), request method (e.g., a dropdown selector providing choices such as “GET” and “POST”), URL string parameters (e.g., text boxes arranged as “name:value” pairs which may permit the selection of extant object 130 parameters 140), and additional settings for the API call (such as limiting the number of simultaneous requests to the API, header data, etc.).

[0156] Upon making an API call, the external system may apply its rules to the submitted query and take action thereupon. For example, the external system may return JSON formatted data to platform 100. An API call node 110 may receive the external system’s response. The node 110 may utilize the external system’s response by, for example, setting parameters 140 (of the object 130 that precipitated the API call) in accordance therewith. In an example, a call to a stock value API may return a dollar value and an associated timestamp. In this example, upon receipt, the calling node 110 may set parameters 140 of the object 130—already equipped with a value for “stock_ticker_symbol” but blank in the parameters 140 “value” and “value_timestamp”—to the returned values.

[0157] As with various types of node 110, a UCE 110 specialized to make API calls may, upon creation, be automatically provided by platform 100 with an error handling loop. Such an error handling loop may comprise a built-in conditional to route errors out of the API call node’s 110 queue 120, a conditional node 110 to handle the type of error generated, one or more exit nodes 110, and a “retry” delay node 110 interconnecting back to the API call node 110.

[0158] FIG. 5 depicts an example interface to platform 100 showing example implementation 500. In example implementation 500, a code execution node 510 is shown. In some senses, code execution may be conceptualized as a specialization, in that, in a particular implementation of platform 100, only certain programming languages and functionality may be made available to a user. In some senses, code execution may be conceptualized as a more generalized case of function 160 types, in that other specializations (calling an API, calling a code versioning repository, delaying, applying conditional logic, etc.) could theoretically be performed by the execution of arbitrary code. In various embodiments, platform 100 may expose variables such as parameters 140 to code execution nodes

510. For example, an object’s 130 parameter 140 “parameter_name” may be made available to a code execution node 510 as “data.parameter_name”.

[0159] FIG. 6 depicts an example interface to platform 100 showing example implementation 600. In example implementation 600, a conditional node 610 is shown. In various embodiments, a conditional node 610 may have an arbitrary number of conditions, each providing an originating interconnection point outbound to a node 110. In various embodiments, conditions may operate on object 130 parameters 140 (e.g., “IF vip==TRUE . . .”), on states resulting from lack of an object 130 (e.g., “IF_error==no_object”), on counters 150 (e.g., “IF c_t>30”), or combinations of the aforementioned.

[0160] In some embodiments, conditional nodes 110 may make only certain operators available in certain combinations. In an example, conditional nodes 110 may make only equality, inequality, greater than, less than, and regular expression matching available as conditions, and combinations of conditions only by AND OR logical operators. Different implementations of platform 100 may make available to conditional nodes 110 different subsets of the universe of comparison operators, logical operators, bitwise operators, regular expressions, etc.

[0161] FIG. 7 depicts an example interface to platform 100 showing example implementation 700. In example implementation 700, wherein a database call node 710 is shown. In an example, platform 100 may have an interface for setting up a database connection. In further consistent embodiments, a database call node 110 may reference a database connection. A user interface to platform 100 may provide an input field for making an actual query to the database in a language such as Structured Query Language (SQL), GraphQL, MongoDB Query Language (MQL), etc.

[0162] f. Analytics, Data Visualization, and Dashboard

[0163] Platform 100 may comprise analytics, data visualization, and dashboard functionality. In various embodiments, such functionality may be realtime as well as historical. In various embodiments, such functionality may be based on metrics (i.e., states of counters 150 and parameters 140) from any UCE 110. For example, a metric may correspond to a counter 150 of the number of objects 130 that have reached a successful end node 110 of a process 170. For another example, a metric may be based on a counter 150 of the number of objects 130 in a queue 120 of a particular node 110 having a parameter 140 that meets a particular criteria (“is_vip” is TRUE, “times_contacted” is less than or equal to 3, etc.).

[0164] In various embodiments, a dashboard may be capable of displaying metrics from multiple processes 170. In various embodiments, a dashboard may be capable of displaying different time ranges. Time ranges may include examples such as realtime, past 10 minutes, past hour, past 24 hours, past week, past year, and custom user-specified ranges. In further consistent embodiments, a dashboard may display metrics over those time ranges. By way of some non-limiting examples, a dashboard may display such metrics as: successful process 170 completions over the past 24 hours, current (realtime) number of users in the queue 120 of a particular node 110, process 170 failures due to API call errors over a particular two week period, etc.

[0165] In various embodiments, data visualizations of a dashboard may be available. Such data visualizations may include examples such as bar charts, pie charts, funnel

charts, etc. Data visualizations may comprise, by way of some non-limiting examples: maps of processes 170 that display realtime or historical data about activity in various nodes 110, charts of the activity in interconnected processes 170, geographic visualizations (e.g., maps of where objects 130 are originating from, charts that show errors at a particular node 110 by country), heat maps, scatter plots, bubble charts, etc.

[0166] An example interface to platform 100, implementing analytics, data visualization, and dashboard functionality, is depicted in FIGS. 8A-C. FIG. 8A depicts an example interface to platform 100 showing example implementation 800, wherein a selection (for inclusion in a dashboard) of metrics is presented for nodes 810 of processes 870. FIG. 8B depicts an example interface to platform 100 showing example implementation 800, wherein dashboard 820 (including chosen metrics from nodes 810) is shown. FIG. 8C depicts an example interface to platform 100 showing example implementation 800, wherein a data visualization 830 of dashboard 820 is shown. While each of the metrics selected in example implementation 800 are from a single process 170, entitled “Tracking”, other processes 170 may have been included on the dashboard and related data visualization. For example, the process 170 “Packages (cities)” seen under processes 870 on FIG. 8A may have been chosen as a source of metrics.

II. Method of Use

[0167] Referring now to FIGS. 9A-C, there is shown an example method 900 of utilizing platform 100 to process inbound customer calls. FIG. 9A depicts the entire process in a flowchart diagram, including two subprocesses 910 and 950 of method 900. FIG. 9B depicts an example interface to platform 100 showing the implementation of subprocess 910 (or the “calls queue”). Example subprocess 910 may comprise a process 170 which may be responsible for maintaining the queue of customer calls (i.e., tracking changes in the state of calls). FIG. 9C depicts an example interface to platform 100 showing the implementation of subprocess 950 (or the “operator queue”). Example subprocess 950 may comprise a process 170 which may be responsible for processing of calls by an operator. Note that the term “queue” as used in “calls queue 910” and “operator queue 950” is in the general sense of processing callers, and not the queue 120 of an individual UCE 110.

[0168] g. Subprocess 910 (the “Calls Queue”)

[0169] Method 900 may begin with a new customer call at stage 911, which may comprise the start node 110 of the calls queue 910. The customer call may be represented by an object 130. Object 130 may proceed to stage 912, which may comprise a node 110 that establishes a queue 120 for use by another process 170. In this example, at stage 912, a new customer call object 130 may be placed into a new customer call queue 120 for use by the operator queue 950. Subprocess 910 may wait for a response from the operator queue 950 (the result of attempting to communicate with the customer) at stage 913.

[0170] Stage 914 may comprise a conditional node 110. In this example, the conditional node 110 may apply the following logic: if the result from the operator queue 950 was a successful communication with the customer, object 130 may proceed to stage 915, which may be an end node 110 of subprocess 910 indicating successful communication. If the result from operator queue 950 was not a successful

communication, object 130 may proceed to stage 916. Stage 916 may comprise a delay node 110 that holds object 130 for 3 hours before proceeding to stage 917.

[0171] Stage 917 may comprise a node 110 that establishes a queue 120 for use by another process 170. In this example, at stage 917, object 130 may be placed into a customer re-call queue 120 for use by the operator queue 950. Subprocess 910 may wait for a response from the operator queue 950 (the result of attempting to communicate with the customer) at stage 918.

[0172] Stage 919 may comprise a conditional node 110. In this example, the conditional node 110 may apply the following logic: if the result from the operator queue 950 was a successful communication with the customer on the re-call, object 130 may proceed to stage 915, which may be an end node 110 of subprocess 910 indicating successful communication. If the result from operator queue 950 was not a successful communication, object 130 may proceed to stage 920.

[0173] Stage 920 may comprise a conditional node 110. In this example, the conditional node 110 may apply the following logic: if object 130 has a call count parameter 140 indicating this was the third attempt to reach the customer, object may proceed to stage 921, which may be an end node 110 of subprocess 910 indicating communications were closed after 3 unsuccessful attempts. If object 130 does not have a call count parameter 140 indicating this was the third attempt to reach the customer, object 130 may proceed to a “Set Parameter” node 110 that sets the call count parameter 140 and on to stage 922 which may comprise the same delay node 110 as stage 916. Object 130 may then proceed (i.e., loop back) to stage 917 to the node 110 that, in this example, establishes the customer re-call queue 120 for use by subprocess 950.

[0174] h. Subprocess 950 (the “Operator Queue”)

[0175] Meanwhile, subprocess 950 may begin at stage 951, which may comprise the start node 110 of the operator queue 950. Stage 952 may comprise a node 110 that attempts to get an object 130 from the (in this example, higher priority) customer re-call queue 120 populated at stage 917.

[0176] Stage 953 may comprise a conditional node 110. In this example, the conditional node 110 may apply the following logic: if the result of stage 952 is that no objects 130 are received (i.e., the customer re-call queue 120 is empty), method 900 may proceed to stage 954. Stage 954 may comprise a node 110 that attempts to get an object 130 from the (in this example, lower priority) new customer call queue 120 populated at stage 912.

[0177] Stage 955, may comprise a conditional node 110. In this example, the conditional node 110 may apply the following logic: if the result of stage 954 is that no objects 130 are received (i.e., the new customer call queue 120 is empty), method 900 may proceed to stage 956, which may be an end node 110 of subprocess 950 indicating there are no customers to contact.

[0178] If the result of either stage 954 or stage 952 is that an object 130 is received, object 130 may proceed (from stage 955 or stage 953, respectively) to stage 957. Stage 957 may comprise a node 110 which may send the object (“task”) 130 to an operator to initiate communication. Subprocess 950 may await the results of that communication (e.g., successful or unsuccessful) at stage 958. Object 130 may proceed to stage 959. Stage 959 may comprise a “Modify Task” node 110 that utilizes the result of the

communication (e.g., successful or unsuccessful) to update an object **130** in subprocess **910** waiting at either stage **913** or stage **918** for a response from the operator queue **950** (for new calls from stage **954** or re-calls from stage **952**, respectively). Subprocess **950** may conclude with an end node **110** at stage **960**.

[0179] The order of stages presented are only illustrative of the possibilities and those steps can be executed or performed in any suitable fashion. Moreover, the various features of the examples described here are not mutually exclusive. Rather any feature of any example described here can be incorporated into any other suitable example. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

III. Task Publish Integration Marketplace

[0180] FIG. **16** illustrates a process flow **1600** of the Task Publish-Integration (TPI) marketplace in accordance with aspects of the platform. In embodiments, a point of input for the TPI marketplace may accept command messages. For example, the command messages may include text commands, selections from one or more menus, and/or other command types. The TPI marketplace may synthesize the command messages into an executable workflow. For example, the process flow **1600** may begin with a task publishing workflow. At Step **1602**, the TPI marketplace may receive, via the point of input, a command to create a new process. The command to create a new process may include, for example, an explicit textual command entered by a user to begin a new process, a selection of a new process from a drop-down menu, or the like, there may be many ways to create a new process.

[0181] Once a new process is created, the TPI marketplace may receive one or more commands to define a list of nodes for inclusion in the process at Step **1604**. As discussed above, defining a node may include, for example, setting a node name, setting one or more parameters of a node, and setting a function of the node. As particular examples, a node function may be to execute a particular API call, to execute a particular code segment, to determine a next node based on one or more conditions, and/or the like. There are many different functions that a node may perform.

[0182] Once the nodes are defined, the TPI marketplace may create the nodes in Step **1606**. Creating the nodes may comprise creating the nodes based on the listed definitions from Step **1604**. In some embodiments, creating the nodes may comprise causing the nodes to be created. For example, creating the nodes may comprise creating code that, when executed, causes a processor to create the nodes.

[0183] At Step **1608**, the TPI marketplace may receive commands to define code to be added to one or more of the created nodes. In some embodiments, defining the code may include entering one or more lines of code. Additionally or alternatively, defining the code may include specifying one or more data files containing executable code. The defined code may comprise code executable by a microprocessor. In some embodiments, the code may be code that must be compiled prior to execution. In other embodiments, the code may comprise code that is already compiled and is ready for execution, or code written in an interpreted language that is not compiled prior to being executed. In still other embodi-

ments, the code may comprise one or more executable files that, when executed, cause a hardware processor to perform particular actions.

[0184] At Step **1610**, the TPI marketplace may add the defined code to the nodes. Adding the code to the nodes may include, for example, inserting the code into the node. Alternatively or additionally, inserting the code may include inserting a reference to the code into the node. For example, a filename of a file containing the code may be inserted into the node.

[0185] At Step **1612**, the TPI marketplace may receive one or more commands to define transitional conditions between the nodes. That is, a user may define one or more conditions that, if satisfied, will cause processing to transition from one node to a next identified node. In some embodiments, the transition logic may include one or more “if” statements. An “if” statement may be a statement that causes a transition from one node to a next identified node if a condition described in the “if” statement is satisfied. In some embodiments, the transition logic may include one or more “go” statements. A “go” statement may be a statement that unconditionally causes a transition from one node to a next identified node. The “if” and “go” statements may be added to the nodes in step **1614**.

[0186] As shown in FIG. **16**, following Step **1614**, the process flow **1600** may proceed to a task checkout workflow, beginning with the TPI marketplace publishing a graph at Step **1616**. Publishing the graph may include making the graph created in the previous Steps **1602-1614** available to be viewed by one or more third parties in the marketplace. In some embodiments, publishing the graph may further include cataloguing the graph and/or the process associated with the graph. In some embodiments, publishing the graph may include allowing users to search for the graph via the marketplace.

[0187] At Step **1618**, a third party entity (e.g., an entity other than the entity responsible for creation of the graph) may access the published graph. Accessing the graph may include, receiving a search result of a marketplace search that includes a reference to the graph. In some embodiments, accessing the graph may include downloading the graph and/or the process associated with the graph. In some embodiments, accessing the graph may comprise completing a financial transaction, whereby the marketplace prevents accessing the graph until completion of the financial transaction.

[0188] Following access of the graph at Step **1618**, in some embodiments, the marketplace may receive request to modify the graph at Step **1620**. In embodiments receiving the request may include receiving a command via a user interface. The request may be received from the third party entity that accessed the graph.

[0189] Responsive to receiving the request, at Step **1622**, the marketplace may cause the third party entity to access a modification interface for modifying the graph and/or the process associated with the graph. The modification interface may take the form of a web portal accessible to the third party entity. In some embodiments, the modification interface may take the form of a stand-alone application.

[0190] The modification interface may allow the third party entity to create a copy of the graph and/or the process associated with the graph. The modification interface may allow the third party entity to make changes to one or more of the nodes that make up the graph. The modification

interface may allow the third party entity to add one or more nodes to the graph. The modification interface may allow the third party entity to remove one or more nodes from the graph. In embodiments, the third party entity may use the modification interface to make one or more changes to the graph and the process associated with the graph, such that the process may be better suited for integration into a workflow of the third party entity.

[0191] At step **1624** the modified graph can be integrated into a process flow of the third party entity. In some embodiments, integrating the modified graph may include downloading the modified graph and/or the process associated with the graph. The modified graph and/or the process associated with the graph may be stored at a device configured to manage the third party entity process flow, and the modified process may be executed by the device.

[0192] Alternatively or additionally, integrating the modified graph may include providing a system configured to manage the third party entity process flow with a reference to the modified graph. For example, the TPI marketplace and/or the modification interface may provide a reference (e.g., a uniform resource locator, a file name, or a path name) associated with the modified graph. The reference may be stored at a device configured to manage the third party entity process flow, and the modified process may be executed by the device.

[0193] As a particular example, in the context of a process for automated invoice payment, the TPI marketplace may create the process based on user input, determining data parameters used for designation of the necessary states enabling the automated invoice payment process. The determined parameters may include, for example, a service type, a customer identifier, and an invoice amount. The system may assemble a data structure and a process flow (e.g., using UCEs as nodes) based on the necessary states. The assembled process flow may pass requests from one node to a next node, following a graphical diagram (graph) of the process and upon occurrence of an event significant for the process. In embodiments, the event may be at least one of the customers presence in the service channel, delivery of the invoice, or confirmed customer consent to pay. Once assembled, the TPI marketplace may publish a process graph associated with the assembled automatic invoice payment process. The graph may be published with a description of the process on a publicly-accessible portal, such as a web site. In some embodiments, the published graph may be one of a plurality of independent graphs. Each graph within the portal may be accessible by a third party end-user. For example, the end user may click a link or otherwise select the published graph associated with the automated invoice payment process to access the graph. In embodiments, the user may further request to modify the graph. Responsive to the request to modify, the TPI marketplace may provide access to a modification interface allowing the end user to modify the graph and the automated invoice payment process associated therewith. The TPI marketplace may further enable the end user to integrate the modified process into the end user process flow, allowing the end user to launch the customized automated invoice payment process from a device configured to manage the end user's process flow, such as a web server associated with the end user.

[0194] In some embodiments, a process may include a certain finite number of nodes and transitions, and may be presented as a graph showing each of the nodes, intercon-

nected with directed edges representing the transitions between nodes. The directed edges may define relationships between nodes and a sequence of a process. The process may be carried out by movement of one or more data objects from an input position (e.g., a first node) to an output position. In embodiments, the data objects may be moved from one node to a next node based on events, which may be generated by interaction with an external system (e.g., interaction with an API) and/or performance of one or more functions in the nodes during process execution.

[0195] In some embodiments, the creation and publishing of processes onto a publicly-accessible marketplace (e.g., the TPI marketplace) may be controlled centrally. For example, a global entity (e.g., the global entity **1520**) may determine whether a local entity (e.g., local entities **1510a**, **1510b**) is permitted to create and/or publish a process. In some embodiments, the decision may be based, at least in part, on one or more features of the entity. Alternatively or additionally, the decision may be based on one or more features of the process.

[0196] In some embodiments the creation and publishing of processes onto a publicly-accessible marketplace (e.g., the TPI marketplace) may be controlled in a distributed manner. For example, a plurality (e.g., all) of the local entities the connect to the marketplace may be configured to moderate the marketplace. A created graph may be placed in a moderator queue, and one or more of the moderator entities may determine whether or not to publish the graph to the marketplace.

[0197] In some embodiments, a single entity (e.g., the global entity) may be configured to catalogue the published processes. For example the cataloguing may include creating a thumbnail image of the published process, and an end-user may browse a page of thumbnail images to select a process. Selection of a thumbnail image may cause the marketplace to present additional cataloguing data, including a category assigned to the process, a of the process, a process description, etc.

[0198] Alternatively, there may not be a single entity configured to catalogue the published processes. Rather, each entity may be responsible for cataloguing any processes created by the entity.

IV. Platform Architecture

[0199] i. Architectural Implementations

[0200] FIG. **10** depicts a general architectural overview of an example implementation **1000** of platform **100**. Example implementation **1000** may include API applications (including acceptance of requests, process management, managing state diagrams, etc.), counters **150**, cache (i.e., memory), main logic applications (including most specialized UCE **110** functionality), HTTP applications (including outgoing API requests), user's code applications (including compilation and execution of arbitrary code as functions **160**), the message queue (labeled "MQ"), and database (representing a storage medium in which platform **100** may be implemented).

[0201] FIG. **11** depicts a general architectural overview of an example implementation **1100** of platform **100**. Example implementation **1100** may largely mirror example implementation **1000** as depicted in FIG. **10**, with a key difference being this example system's **100** implementation in memory versus in a storage medium (compare positions of "Cache" and "Database" in FIG. **10** with positions of "DB", which

stands for “database”, and “Cache” in FIG. 11). Implementation of platform 100 in memory may have desirable effects such as enhanced efficiency and faster launch speed.

[0202] FIG. 12 depicts a simplified activity diagram of the interpretation algorithm 1200 that may run at each node 110 in an example implementation of platform 100. In various embodiments, the manner in which UCEs 110 may implement functions 160, check counters 150, and process objects 130 may be conceptualized in accordance with the simplified activity diagram of FIG. 12. The diagram is simplified in at least the following ways: not all erroneous situations and error handling points are depicted, not all functions 160 are depicted, details related to the transformation of data formats in the organization of external API calls are not depicted, the features of operation in start and end nodes 110 are not depicted, not all object 130 routing is depicted, etc.

[0203] An example implementation 1300 of platform 100 demonstrating Turing completeness is presented in FIGS. 13A-B. FIG. 13A depicts a block diagram of example implementation 1300, while FIG. 13B depicts an example interface to platform 100 showing example implementation 1300.

[0204] FIG. 15 illustrates an operating environment 1500 showing interactions between local entities 1510a and 1510b, such as companies or other end users, employing the UCE-based finite-state tasks, and a global entity 1520. While FIG. 13 illustrates an operating environment 1500 including two local entities 1510, those of skill in the art will recognize that an operating environment may include more or fewer entities. In embodiments the local entities 1510a and 1510b may communicate with one another and/or with the global entity 1520. In some embodiments communication may include establishing a communication link, such as a direct or indirect wired or wireless connection, between the entities. In some embodiments, the local entities 1510a and 1510b may communicate with any number of other local entities using a standard protocol, without the need of an API. The limitations of the API are that there are disparate formats making communication complex.

[0205] In some embodiments, the communication among the entities 1510, 1520 in the operating environment 1500 may include peer-to-peer communication. That is, peer entities (e.g., local entities 1510a and 1510b) may communicate directly with one another. For example, the entities may establish a communication link (e.g., link 1-2) to communicate data, including, for example, tasks or processes, algorithms, or other information among themselves.

[0206] In some embodiments, the communication among the entities 1510, 1520 in the operating environment 1500 may include algorithm hosting. That is, a global entity 1520 may include a marketplace 1522. The marketplace 1522 may store one or more algorithms for retrieval by third parties. Each entity 1510a, 1510b may access the marketplace 1522 to publish and/or retrieve algorithms to the marketplace. For example, communication link 3-8 shows a connection between the global entity 1520 and the local entity 1510a; and communication link 3-4 shows a connection between the global entity and the local entity 1510b. In embodiments the communication links 3-8 and 3-4 may be used to publish algorithms from the local entity to the marketplace 1522 of the global entity 1520, and/or to retrieve algorithms from the global entity to the local entity.

[0207] In some embodiments, the communication among the entities 1510, 1520 in the operating environment 1500

may include process hosting. That is, the global entity 1520 may include a marketplace 1522 that may store one or more processes for retrieval by third parties. Each entity 1510a, 1510b may access the marketplace 1522 to publish and/or retrieve processes to the marketplace. For example, communication link 6-7 shows a connection between the global entity 1520 and the local entity 1510a; and communication link 5-6 shows a connection between the global entity and the local entity 1510b. In embodiments the communication links 5-6 and 6-7 may be used to publish processes from the local entity to the marketplace 1522 of the global entity 1520, and/or to retrieve processes from the global entity to the local entity.

[0208] In some embodiments, the communication among the entities 1510, 1520 in the operating environment 1500 may include multiple features, as discussed above. For example, the communication may include peer-to-peer connection with algorithm hosting; peer-to-peer connection with process hosting; peer-to-peer connection with algorithm hosting and process hosting; algorithm hosting without peer-to-peer connection; process hosting without peer-to-peer hosting; and/or algorithm and process hosting without peer-to-peer connection.

[0209] In some embodiments, the global entity 1520 may store limits 1524 indicating restrictions as to how each of the processes and/or algorithms in the marketplace 1522 may be used. For example, the limits may restrict, for each process and/or algorithm stored in the marketplace, which entities can access the process or algorithm. As particular examples, the limits may specify that a process or algorithm can only be used by a particular business type (e.g., a non-profit business), a particular size of business (e.g., a business having less than a defined number of employees and/or a business with revenue below a defined revenue threshold), and/or any other limitations on use the publishing entity wishes to place on the algorithm and/or process.

[0210] In some embodiments, the global entity 1520 may include a billing unit 1526. There may be a fee associated with a local entity 1510 publishing a process or algorithm to the marketplace 1522 and/or retrieving a process or algorithm from the marketplace. In some embodiments, the fee may be collected by the global entity. When a fee is collected for retrieving a process or algorithm from the marketplace 1522, the collected fee may be split between the global entity 1520 and the end user that published the process or algorithm to the global entity. Alternatively, the global entity 1520 may retain the entire fee.

[0211] As a particular example, the operating environment (e.g., operating environment 1500) may include two local entities (e.g., local entities 1510a and 1510b) and a global entity (e.g., global entity 1520) having a marketplace (e.g., marketplace 1522) that stores a plurality of processes and/or algorithms. Communication links (e.g., links 5-6, 6-7) may be established between the global entity and the two local entities. In this way, the local entities may be able to access the marketplace hosted on the global entity to search processes hosted in the marketplace and/or to publish processes to the marketplace. Similarly, communication links (e.g., links 3-8, 3-4) may be established between the global entity and the local entities to allow the local entities to search algorithms hosted in the marketplace and/or to publish algorithms to the marketplace. In embodiments, each links access to the process in a marketplace by the local entities with established limits. For example, each communication

link may connects the local entity to the marketplace with limitations on what the local entity may view based on parameters associated with the entity. For example, each entity may only be able to view algorithms and/or processes the entity is permitted to download (e.g., algorithms and/or processes for which the parameters of the entity meet the limitations of the algorithm or process).

[0212] A local entity may retrieve (e.g., copy) an algorithm or process from the global entity to the local entity via the associate communication link.

[0213] In some embodiments, a local entity may have direct access to the marketplace. That is, communication (e.g., data exchange in terms of nodal-based data transfer) may be achieved using a standardized and format-agnostic protocol. Alternatively, communication may be achieved via an API interface.

[0214] The marketplace 1522 may be a resource with a catalog of voluntarily published limited and/or publicly available resources (e.g., algorithms and/or processes) provided by one or more local entities. Visually, the marketplace 1522 may depict each process or algorithm as a graph, where vertices of the graph correspond to nodes (UCEs) connected by directed edges. In some embodiments, once an algorithm or process has been retrieved (e.g., copied) by a local entity, the local entity may modify the process or algorithm. The local entity may execute the process or algorithm, including any modifications, by launching the node-based finite-state task loop associated with the process or algorithm. In embodiments, the node-based finite state task loop may be formed by a plurality of UCEs, as discussed above.

[0215] j. System Implementations

[0216] Embodiments of platform 100 may comprise aspects including, but not limited to, mobile software applications (or “apps”), websites, web applications, desktop software, server software, embedded software, microcontrollers, databases, wired and wireless networking hardware and software, and various computing devices. Moreover, platform 100 or aspects thereof may be hosted one or more physical or virtual servers, cloud computing services, blockchain platforms, or distributed computing platforms. Alternatively, or in addition, platform 100 may be implemented in one or more of a plurality of computing devices.

[0217] Embodiments of the present disclosure may comprise a system having a central processing unit (CPU) 1420, a bus 1430, a memory unit 1440, a power supply unit (PSU) 1450, and one or more Input/Output (I/O) units. The CPU 1420 coupled to the memory unit 1440 and the plurality of I/O units 1460 via the bus 1430, all of which are powered by the PSU 1450. It should be understood that, in some embodiments, each disclosed unit may actually be a plurality of such units for the purposes of redundancy, high availability, and/or performance. The combination of the presently disclosed units is configured to perform the stages any method disclosed herein.

[0218] FIG. 14 is a block diagram of a system including computing device 1400. Consistent with an embodiment of the disclosure, the aforementioned CPU 1420, the bus 1430, the memory unit 1440, a PSU 1450, and the plurality of I/O units 1460 may be implemented in a computing device, such as computing device 1400 of FIG. 14. Any suitable combination of hardware, software, or firmware may be used to implement the aforementioned units. For example, the CPU 1420, the bus 1430, and the memory unit 1440 may be

implemented with computing device 1400 or any of other computing devices 1400, in combination with computing device 1400. The aforementioned system, device, and components are examples and other systems, devices, and components may comprise the aforementioned CPU 1420, the bus 1430, the memory unit 1440, consistent with embodiments of the disclosure.

[0219] At least one computing device 1400 may be embodied as any of the computing elements illustrated in all of the attached figures. A computing device 1400 does not need to be electronic, nor even have a CPU 1420, nor bus 1430, nor memory unit 1440. The definition of the computing device 1400 to a person having ordinary skill in the art is “A device that computes, especially a programmable electronic machine that performs high-speed mathematical or logical operations or that assembles, stores, correlates, or otherwise processes information.” Any device which processes information qualifies as a computing device 1400, especially if the processing is purposeful.

[0220] With reference to FIG. 14, a system consistent with an embodiment of the disclosure may include a computing device, such as computing device 1400. In a basic configuration, computing device 1400 may include at least one clock module 1410, at least one CPU 1420, at least one bus 1430, and at least one memory unit 1440, at least one PSU 1450, and at least one I/O 1460 module, wherein I/O module may be comprised of, but not limited to a non-volatile storage sub-module 1461, a communication sub-module 1462, a sensors sub-module 1463, and a peripherals sub-module 1464.

[0221] A system consistent with an embodiment of the disclosure the computing device 1400 may include the clock module 1410 may be known to a person having ordinary skill in the art as a clock generator, which produces clock signals. Clock signal is a particular type of signal that oscillates between a high and a low state and is used like a metronome to coordinate actions of digital circuits. Most integrated circuits (ICs) of sufficient complexity use a clock signal in order to synchronize different parts of the circuit, cycling at a rate slower than the worst-case internal propagation delays. The preeminent example of the aforementioned integrated circuit is the CPU 1420, the central component of modern computers, which relies on a clock. The only exceptions are asynchronous circuits such as asynchronous CPUs. The clock 1410 can comprise a plurality of embodiments, such as, but not limited to, single-phase clock which transmits all clock signals on effectively 1 wire, two-phase clock which distributes clock signals on two wires, each with non-overlapping pulses, and four-phase clock which distributes clock signals on 4 wires.

[0222] Many computing devices 1400 use a “clock multiplier” which multiplies a lower frequency external clock to the appropriate clock rate of the CPU 1420. This may allow the CPU 1420 to operate at a much higher frequency than the rest of the computer, which affords performance gains in situations where the CPU 1420 does not need to wait on an external factor (like memory 1440 or input/output 1460). Some embodiments of the clock 1410 may include dynamic frequency change, where, the time between clock edges can vary widely from one edge to the next and back again.

[0223] A system consistent with an embodiment of the disclosure the computing device 1400 may include the CPU unit 1420 comprising at least one CPU Core 1421. A plurality of CPU cores 1421 may comprise identical CPU

cores **1421**, such as, but not limited to, homogeneous multi-core systems. It is also possible for the plurality of CPU cores **1421** to comprise different CPU cores **1421**, such as, but not limited to, heterogeneous multi-core systems, big.LITTLE systems and some AMD accelerated processing units (APU). The CPU unit **1420** reads and executes program instructions which may be used across many application domains, for example, but not limited to, general purpose computing, embedded computing, network computing, digital signal processing (DSP), and graphics processing (GPU). The CPU unit **1420** may run multiple instructions on separate CPU cores **1421** at the same time. The CPU unit **1420** may be integrated into at least one of a single integrated circuit die and multiple dies in a single chip package. The single integrated circuit die and multiple dies in a single chip package may contain a plurality of other aspects of the computing device **1400**, for example, but not limited to, the clock **1410**, the CPU **1420**, the bus **1430**, the memory **1440**, and I/O **1460**.

[0224] The CPU unit **1420** may contain cache **1422** such as, but not limited to, a level 1 cache, level 2 cache, level 3 cache or combination thereof. The aforementioned cache **1422** may or may not be shared amongst a plurality of CPU cores **1421**. The cache **1422** sharing comprises at least one of message passing and inter-core communication methods may be used for the at least one CPU Core **1421** to communicate with the cache **1422**. The inter-core communication methods may comprise, but not limited to, bus, ring, two-dimensional mesh, and crossbar. The aforementioned CPU unit **1420** may employ symmetric multiprocessing (SMP) design.

[0225] The plurality of the aforementioned CPU cores **1421** may comprise soft microprocessor cores on a single field programmable gate array (FPGA), such as semiconductor intellectual property cores (IP Core). The plurality of CPU cores **1421** architecture may be based on at least one of, but not limited to, Complex instruction set computing (CISC), Zero instruction set computing (ZISC), and Reduced instruction set computing (RISC). At least one of the performance-enhancing methods may be employed by the plurality of the CPU cores **1421**, for example, but not limited to Instruction-level parallelism (ILP) such as, but not limited to, superscalar pipelining, and Thread-level parallelism (TLP).

[0226] Consistent with the embodiments of the present disclosure, the aforementioned computing device **1400** may employ a communication system that transfers data between components inside the aforementioned computing device **1400**, and/or the plurality of computing devices **1400**. The aforementioned communication system will be known to a person having ordinary skill in the art as a bus **1430**. The bus **1430** may embody internal and/or external plurality of hardware and software components, for example, but not limited to a wire, optical fiber, communication protocols, and any physical arrangement that provides the same logical function as a parallel electrical bus. The bus **1430** may comprise at least one of, but not limited to a parallel bus, wherein the parallel bus carry data words in parallel on multiple wires, and a serial bus, wherein the serial bus carry data in bit-serial form. The bus **1430** may embody a plurality of topologies, for example, but not limited to, a multidrop/electrical parallel topology, a daisy chain topology, and a

connected by switched hubs, such as USB bus. The bus **1430** may comprise a plurality of embodiments, for example, but not limited to:

- [0227] Internal data bus (data bus) **1431**/Memory bus
- [0228] Control bus **1432**
- [0229] Address bus **1433**
- [0230] System Management Bus (SMBus)
- [0231] Front-Side-Bus (FSB)
- [0232] External Bus Interface (EBI)
- [0233] Local bus
- [0234] Expansion bus
- [0235] Lightning bus
- [0236] Controller Area Network (CAN bus)
- [0237] Camera Link
- [0238] ExpressCard
- [0239] Advanced Technology management Attachment (ATA), including embodiments and derivatives such as, but not limited to, Integrated Drive Electronics (IDE)/Enhanced IDE (EIDE), ATA Packet Interface (ATAPI), Ultra-Direct Memory Access (UDMA), Ultra ATA (UATA)/Parallel ATA (PATA)/Serial ATA (SATA), CompactFlash (CF) interface, Consumer Electronics ATA (CE-ATA)/Fiber Attached Technology Adapted (FATA), Advanced Host Controller Interface (AHCI), SATA Express (SATAe)/External SATA (eSATA), including the powered embodiment eSATAp/Mini-SATA (mSATA), and Next Generation Form Factor (NGFF)/M.2.
- [0240] Small Computer System Interface (SCSI)/Serial Attached SCSI (SAS)
- [0241] HyperTransport
- [0242] InfiniBand
- [0243] RapidIO
- [0244] Mobile Industry Processor Interface (MIPI)
- [0245] Coherent Processor Interface (CAPI)
- [0246] Plug-n-play
- [0247] 1-Wire
- [0248] Peripheral Component Interconnect (PCI), including embodiments such as, but not limited to, Accelerated Graphics Port (AGP), Peripheral Component Interconnect eXtended (PCI-X), Peripheral Component Interconnect Express (PCI-e) (e.g., PCI Express Mini Card, PCI Express M.2 [Mini PCIe v2], PCI Express External Cabling [ePCIe], and PCI Express OCuLink [Optical Copper{Cu} Link]), Express Card, AdvancedTCA, AMC, Universal 10, Thunderbolt/Mini DisplayPort, Mobile PCIe (M-PCIe), U.2, and Non-Volatile Memory Express (NVMe)/Non-Volatile Memory Host Controller Interface Specification (NVMHCIS).
- [0249] Industry Standard Architecture (ISA), including embodiments such as, but not limited to Extended ISA (EISA), PC/XT-bus/PC/AT-bus/PC/104 bus (e.g., PC/104-Plus, PCI/104-Express, PCI/104, and PCI-104), and Low Pin Count (LPC).
- [0250] Music Instrument Digital Interface (MIDI)
- [0251] Universal Serial Bus (USB), including embodiments such as, but not limited to, Media Transfer Protocol (MTP)/Mobile High-Definition Link (MHL), Device Firmware Upgrade (DFU), wireless USB, Inter-Chip USB, IEEE 1394 Interface/Firewire, Thunderbolt, and eXtensible Host Controller Interface (xHCI).
- [0252] Consistent with the embodiments of the present disclosure, the aforementioned computing device **1400** may

employ hardware integrated circuits that store information for immediate use in the computing device **1400**, know to the person having ordinary skill in the art as primary storage or memory **1440**. The memory **1440** operates at high speed, distinguishing it from the non-volatile storage sub-module **1461**, which may be referred to as secondary or tertiary storage, which provides slow-to-access information but offers higher capacities at lower cost. The contents contained in memory **1440**, may be transferred to secondary storage via techniques such as, but not limited to, virtual memory and swap. The memory **1440** may be associated with addressable semiconductor memory, such as integrated circuits consisting of silicon-based transistors, used for example as primary storage but also other purposes in the computing device **1400**. The memory **1440** may comprise a plurality of embodiments, such as, but not limited to volatile memory, non-volatile memory, and semi-volatile memory. It should be understood by a person having ordinary skill in the art that the ensuing are non-limiting examples of the aforementioned memory:

- [0253] Volatile memory which requires power to maintain stored information, for example, but not limited to, Dynamic Random-Access Memory (DRAM) **1441**, Static Random-Access Memory (SRAM) **1442**, CPU Cache memory **1425**, Advanced Random-Access Memory (A-RAM), and other types of primary storage such as Random-Access Memory (RAM).
- [0254] Non-volatile memory which can retain stored information even after power is removed, for example, but not limited to, Read-Only Memory (ROM) **1443**, Programmable ROM (PROM) **1444**, Erasable PROM (EPROM) **1445**, Electrically Erasable PROM (EEPROM) **1446** (e.g., flash memory and Electrically Alterable PROM [EAPROM]), Mask ROM (MROM), One Time Programmable (OTP) ROM/Write Once Read Many (WORM), Ferroelectric RAM (FeRAM), Parallel Random-Access Machine (PRAM), Split-Transfer Torque RAM (STT-RAM), Silicon Oxime Nitride Oxide Silicon (SONOS), Resistive RAM (RRAM), Nano RAM (NRAM), 3D XPoint, Domain-Wall Memory (DWM), and millipede memory.
- [0255] Semi-volatile memory which may have some limited non-volatile duration after power is removed but loses data after said duration has passed. Semi-volatile memory provides high performance, durability, and other valuable characteristics typically associated with volatile memory, while providing some benefits of true non-volatile memory. The semi-volatile memory may comprise volatile and non-volatile memory and/or volatile memory with battery to provide power after power is removed. The semi-volatile memory may comprise, but not limited to spin-transfer torque RAM (STT-RAM).

[0256] Consistent with the embodiments of the present disclosure, the aforementioned computing device **1400** may employ the communication system between an information processing system, such as the computing device **1400**, and the outside world, for example, but not limited to, human, environment, and another computing device **1400**. The aforementioned communication system will be known to a person having ordinary skill in the art as I/O **1460**. The I/O module **1460** regulates a plurality of inputs and outputs with regard to the computing device **1400**, wherein the inputs are a plurality of signals and data received by the computing

device **1400**, and the outputs are the plurality of signals and data sent from the computing device **1400**. The I/O module **1460** interfaces a plurality of hardware, such as, but not limited to, non-volatile storage **1461**, communication devices **1462**, sensors **1463**, and peripherals **1464**. The plurality of hardware is used by the at least one of, but not limited to, human, environment, and another computing device **1400** to communicate with the present computing device **1400**. The I/O module **1460** may comprise a plurality of forms, for example, but not limited to channel I/O, port mapped I/O, asynchronous I/O, and Direct Memory Access (DMA).

[0257] Consistent with the embodiments of the present disclosure, the aforementioned computing device **1400** may employ the non-volatile storage sub-module **1461**, which may be referred to by a person having ordinary skill in the art as one of secondary storage, external memory, tertiary storage, off-line storage, and auxiliary storage. The non-volatile storage sub-module **1461** may not be accessed directly by the CPU **1420** without using intermediate area in the memory **1440**. The non-volatile storage sub-module **1461** does not lose data when power is removed and may be two orders of magnitude less costly than storage used in memory module, at the expense of speed and latency. The non-volatile storage sub-module **1461** may comprise a plurality of forms, such as, but not limited to, Direct Attached Storage (DAS), Network Attached Storage (NAS), Storage Area Network (SAN), nearline storage, Massive Array of Idle Disks (MAID), Redundant Array of Independent Disks (RAID), device mirroring, off-line storage, and robotic storage. The non-volatile storage sub-module (**1461**) may comprise a plurality of embodiments, such as, but not limited to:

- [0258] Optical storage, for example, but not limited to, Compact Disk (CD) (CD-ROM/CD-R/CD-RW), Digital Versatile Disk (DVD) (DVD-ROM/DVD-R/DVD+R/DVD-RW/DVD+RW/DVD±RW/DVD+R DL/DVD-RAM/HD-DVD), Blu-ray Disk (BD) (BD-ROM/BD-R/BD-RE/BD-R DL/BD-RE DL), and Ultra-Density Optical (UDO).
- [0259] Semiconductor storage, for example, but not limited to, flash memory, such as, but not limited to, USB flash drive, Memory card, Subscriber Identity Module (SIM) card, Secure Digital (SD) card, Smart Card, CompactFlash (CF) card, Solid-State Drive (SSD) and memristor.
- [0260] Magnetic storage such as, but not limited to, Hard Disk Drive (HDD), tape drive, carousel memory, and Card Random-Access Memory (CRAM).
- [0261] Phase-change memory
- [0262] Holographic data storage such as Holographic Versatile Disk (HVD)
- [0263] Molecular Memory
- [0264] Deoxyribonucleic Acid (DNA) digital data storage

[0265] Consistent with the embodiments of the present disclosure, the aforementioned computing device **1400** may employ the communication sub-module **1462** as a subset of the I/O **1460**, which may be referred to by a person having ordinary skill in the art as at least one of, but not limited to, computer network, data network, and network. The network allows computing devices **1400** to exchange data using connections, which may be known to a person having ordinary skill in the art as data links, between network nodes. The nodes comprise network computer devices **1400**

that originate, route, and terminate data. The nodes are identified by network addresses and can include a plurality of hosts consistent with the embodiments of a computing device **1400**. The aforementioned embodiments include, but not limited to personal computers, phones, servers, drones, and networking devices such as, but not limited to, hubs, switches, routers, modems, and firewalls.

[0266] Two nodes can be said are networked together, when one computing device **1400** is able to exchange information with the other computing device **1400**, whether or not they have a direct connection with each other. The communication sub-module **1462** supports a plurality of applications and services, such as, but not limited to World Wide Web (WWW), digital video and audio, shared use of application and storage computing devices **1400**, printers/scanners/fax machines, email/online chat/instant messaging, remote control, distributed computing, etc. The network may comprise a plurality of transmission mediums, such as, but not limited to conductive wire, fiber optics, and wireless. The network may comprise a plurality of communications protocols to organize network traffic, wherein application-specific communications protocols are layered, may be known to a person having ordinary skill in the art as carried as payload, over other more general communications protocols. The plurality of communications protocols may comprise, but not limited to, IEEE 802, ethernet, Wireless LAN (WLAN/Wi-Fi), Internet Protocol (IP) suite (e.g., TCP/IP, UDP, Internet Protocol version 4 [IPv4], and Internet Protocol version 6 [IPv6]), Synchronous Optical Networking (SONET)/Synchronous Digital Hierarchy (SDH), Asynchronous Transfer Mode (ATM), and cellular standards (e.g., Global System for Mobile Communications [GSM], General Packet Radio Service [GPRS], Code-Division Multiple Access [CDMA], and Integrated Digital Enhanced Network [IDEN]).

[0267] The communication sub-module **1462** may comprise a plurality of size, topology, traffic control mechanism and organizational intent. The communication sub-module **1462** may comprise a plurality of embodiments, such as, but not limited to:

[0268] Wired communications, such as, but not limited to, coaxial cable, phone lines, twisted pair cables (ethernet), and InfiniBand.

[0269] Wireless communications, such as, but not limited to, communications satellites, cellular systems, radio frequency/spread spectrum technologies, IEEE 802.11 Wi-Fi, Bluetooth, NFC, free-space optical communications, terrestrial microwave, and Infrared (IR) communications. Wherein cellular systems embody technologies such as, but not limited to, 3G, 4G (such as WiMax and LTE), and 5G (short and long wavelength).

[0270] Parallel communications, such as, but not limited to, LPT ports.

[0271] Serial communications, such as, but not limited to, RS-232 and USB.

[0272] Fiber Optic communications, such as, but not limited to, Single-mode optical fiber (SMF) and Multi-mode optical fiber (MMF).

[0273] Power Line communications

[0274] The aforementioned network may comprise a plurality of layouts, such as, but not limited to, bus network such as ethernet, star network such as Wi-Fi, ring network, mesh network, fully connected network, and tree network. The network can be characterized by its physical capacity or

its organizational purpose. Use of the network, including user authorization and access rights, differ accordingly. The characterization may include, but not limited to nanoscale network, Personal Area Network (PAN), Local Area Network (LAN), Home Area Network (HAN), Storage Area Network (SAN), Campus Area Network (CAN), backbone network, Metropolitan Area Network (MAN), Wide Area Network (WAN), enterprise private network, Virtual Private Network (VPN), and Global Area Network (GAN).

[0275] Consistent with the embodiments of the present disclosure, the aforementioned computing device **1400** may employ the sensors sub-module **1463** as a subset of the I/O **1460**. The sensors sub-module **1463** comprises at least one of the devices, modules, and subsystems whose purpose is to detect events or changes in its environment and send the information to the computing device **1400**. Sensors are sensitive to the measured property, are not sensitive to any property not measured, but may be encountered in its application, and do not significantly influence the measured property. The sensors sub-module **1463** may comprise a plurality of digital devices and analog devices, wherein if an analog device is used, an Analog to Digital (A-to-D) converter must be employed to interface the said device with the computing device **1400**. The sensors may be subject to a plurality of deviations that limit sensor accuracy. The sensors sub-module **1463** may comprise a plurality of embodiments, such as, but not limited to, chemical sensors, automotive sensors, acoustic/sound/vibration sensors, electric current/electric potential/magnetic/radio sensors, environmental/weather/moisture/humidity sensors, flow/fluid velocity sensors, ionizing radiation/particle sensors, navigation sensors, position/angle/displacement/distance/speed/acceleration sensors, imaging/optical/light sensors, pressure sensors, force/density/level sensors, thermal/temperature sensors, and proximity/presence sensors. It should be understood by a person having ordinary skill in the art that the ensuing are non-limiting examples of the aforementioned sensors:

[0276] Chemical sensors, such as, but not limited to, breathalyzer, carbon dioxide sensor, carbon monoxide/smoke detector, catalytic bead sensor, chemical field-effect transistor, chemiresistor, electrochemical gas sensor, electronic nose, electrolyte-insulator-semiconductor sensor, energy-dispersive X-ray spectroscopy, fluorescent chloride sensors, holographic sensor, hydrocarbon dew point analyzer, hydrogen sensor, hydrogen sulfide sensor, infrared point sensor, ion-selective electrode, nondispersive infrared sensor, microwave chemistry sensor, nitrogen oxide sensor, olfactometer, optode, oxygen sensor, ozone monitor, pellistor, pH glass electrode, potentiometric sensor, redox electrode, zinc oxide nanorod sensor, and biosensors (such as nanosensors).

[0277] Automotive sensors, such as, but not limited to, air flow meter/mass airflow sensor, air-fuel ratio meter, AFR sensor, blind spot monitor, engine coolant/exhaust gas/cylinder head/transmission fluid temperature sensor, hall effect sensor, wheel/automatic transmission/turbine/vehicle speed sensor, airbag sensors, brake fluid/engine crankcase/fuel/oil/tire pressure sensor, camshaft/crankshaft/throttle position sensor, fuel/oil level sensor, knock sensor, light sensor, MAP sensor,

- oxygen sensor (02), parking sensor, radar sensor, torque sensor, variable reluctance sensor, and water-in-fuel sensor.
- [0278] Acoustic, sound and vibration sensors, such as, but not limited to, microphone, lace sensor (guitar pickup), seismometer, sound locator, geophone, and hydrophone.
- [0279] Electric current, electric potential, magnetic, and radio sensors, such as, but not limited to, current sensor, Daly detector, electroscope, electron multiplier, faraday cup, galvanometer, hall effect sensor, hall probe, magnetic anomaly detector, magnetometer, magnetoresistance, MEMS magnetic field sensor, metal detector, planar hall sensor, radio direction finder, and voltage detector.
- [0280] Environmental, weather, moisture, and humidity sensors, such as, but not limited to, actinometer, air pollution sensor, bedwetting alarm, ceilometer, dew warning, electrochemical gas sensor, fish counter, frequency domain sensor, gas detector, hook gauge evaporimeter, humistor, hygrometer, leaf sensor, lysimeter, pyranometer, pyrgeometer, psychrometer, rain gauge, rain sensor, seismometers, SNOTEL, snow gauge, soil moisture sensor, stream gauge, and tide gauge.
- [0281] Flow and fluid velocity sensors, such as, but not limited to, air flow meter, anemometer, flow sensor, gas meter, mass flow sensor, and water meter.
- [0282] Ionizing radiation and particle sensors, such as, but not limited to, cloud chamber, Geiger counter, Geiger-Muller tube, ionization chamber, neutron detection, proportional counter, scintillation counter, semiconductor detector, and thermoluminescent dosimeter.
- [0283] Navigation sensors, such as, but not limited to, air speed indicator, altimeter, attitude indicator, depth gauge, fluxgate compass, gyroscope, inertial navigation system, inertial reference unit, magnetic compass, MHD sensor, ring laser gyroscope, turn coordinator, variometer, vibrating structure gyroscope, and yaw rate sensor.
- [0284] Position, angle, displacement, distance, speed, and acceleration sensors, such as, but not limited to, accelerometer, displacement sensor, flex sensor, free fall sensor, gravimeter, impact sensor, laser rangefinder, LIDAR, odometer, photoelectric sensor, position sensor such as, but not limited to, GPS or Glonass, angular rate sensor, shock detector, ultrasonic sensor, tilt sensor, tachometer, ultra-wideband radar, variable reluctance sensor, and velocity receiver.
- [0285] Imaging, optical and light sensors, such as, but not limited to, CMOS sensor, colorimeter, contact image sensor, electro-optical sensor, infra-red sensor, kinetic inductance detector, LED as light sensor, light-addressable potentiometric sensor, Nichols radiometer, fiber-optic sensors, optical position sensor, thermopile laser sensor, photodetector, photodiode, photomultiplier tubes, phototransistor, photoelectric sensor, photoionization detector, photomultiplier, photoresistor, photoswitch, phototube, scintillometer, Shack-Hartmann, single-photon avalanche diode, superconducting nanowire single-photon detector, transition edge sensor, visible light photon counter, and wavefront sensor.
- [0286] Pressure sensors, such as, but not limited to, barograph, barometer, boost gauge, bourdon gauge, hot filament ionization gauge, ionization gauge, McLeod gauge, Oscillating U-tube, permanent downhole gauge, piezometer, Pirani gauge, pressure sensor, pressure gauge, tactile sensor, and time pressure gauge.
- [0287] Force, Density, and Level sensors, such as, but not limited to, bhangmeter, hydrometer, force gauge or force sensor, level sensor, load cell, magnetic level or nuclear density sensor or strain gauge, piezocapacitive pressure sensor, piezoelectric sensor, torque sensor, and viscometer.
- [0288] Thermal and temperature sensors, such as, but not limited to, bolometer, bimetallic strip, calorimeter, exhaust gas temperature gauge, flame detection/pyrometer, Gardon gauge, Golay cell, heat flux sensor, microbolometer, microwave radiometer, net radiometer, infrared/quartz/resistance thermometer, silicon bandgap temperature sensor, thermistor, and thermocouple.
- [0289] Proximity and presence sensors, such as, but not limited to, alarm sensor, doppler radar, motion detector, occupancy sensor, proximity sensor, passive infrared sensor, reed switch, stud finder, triangulation sensor, touch switch, and wired glove.
- [0290] Consistent with the embodiments of the present disclosure, the aforementioned computing device 1400 may employ the peripherals sub-module 1464 as a subset of the I/O 1460. The peripheral sub-module 1464 comprises ancillary devices uses to put information into and get information out of the computing device 1400. There are 3 categories of devices comprising the peripheral sub-module 1464, which exist based on their relationship with the computing device 1400, input devices, output devices, and input/output devices. Input devices send at least one of data and instructions to the computing device 1400. Input devices can be categorized based on, but not limited to:
- [0291] Modality of input, such as, but not limited to, mechanical motion, audio, visual, and tactile.
- [0292] Whether the input is discrete, such as but not limited to, pressing a key, or continuous such as, but not limited to position of a mouse.
- [0293] The number of degrees of freedom involved, such as, but not limited to, two-dimensional mice vs three-dimensional mice used for Computer-Aided Design (CAD) applications.
- [0294] Output devices provide output from the computing device 1400. Output devices convert electronically generated information into a form that can be presented to humans. Input/output devices perform that perform both input and output functions. It should be understood by a person having ordinary skill in the art that the ensuing are non-limiting embodiments of the aforementioned peripheral sub-module 1464:
- [0295] Input Devices
- [0296] Human Interface Devices (HID), such as, but not limited to, pointing device (e.g., mouse, touchpad, joystick, touchscreen, game controller/gamepad, remote, light pen, light gun, Wii remote, jog dial, shuttle, and knob), keyboard, graphics tablet, digital pen, gesture recognition devices, magnetic ink character recognition, Sip-and-Puff (SNP) device, and Language Acquisition Device (LAD).
- [0297] High degree of freedom devices, that require up to six degrees of freedom such as, but not limited to, camera gimbals, Cave Automatic Virtual Environment (CAVE), and virtual reality systems.

- [0298]** Video Input devices are used to digitize images or video from the outside world into the computing device **1400**. The information can be stored in a multitude of formats depending on the user's requirement. Examples of types of video input devices include, but not limited to, digital camera, digital camcorder, portable media player, webcam, Microsoft Kinect, image scanner, fingerprint scanner, barcode reader, 3D scanner, laser rangefinder, eye gaze tracker, computed tomography, magnetic resonance imaging, positron emission tomography, medical ultrasonography, TV tuner, and iris scanner.
- [0299]** Audio input devices are used to capture sound. In some cases, an audio output device can be used as an input device, in order to capture produced sound. Audio input devices allow a user to send audio signals to the computing device **1400** for at least one of processing, recording, and carrying out commands. Devices such as microphones allow users to speak to the computer in order to record a voice message or navigate software. Aside from recording, audio input devices are also used with speech recognition software. Examples of types of audio input devices include, but not limited to microphone, Musical Instrumental Digital Interface (MIDI) devices such as, but not limited to a keyboard, and headset.
- [0300]** Data Acquisition (DAQ) devices convert at least one of analog signals and physical parameters to digital values for processing by the computing device **1400**. Examples of DAQ devices may include, but not limited to, Analog to Digital Converter (ADC), data logger, signal conditioning circuitry, multiplexer, and Time to Digital Converter (TDC).
- [0301]** Output Devices may further comprise, but not be limited to:
- [0302]** Display devices, which convert electrical information into visual form, such as, but not limited to, monitor, TV, projector, and Computer Output Microfilm (COM). Display devices can use a plurality of underlying technologies, such as, but not limited to, Cathode-Ray Tube (CRT), Thin-Film Transistor (TFT), Liquid Crystal Display (LCD), Organic Light-Emitting Diode (OLED), MicroLED, E Ink Display (ePaper) and Refreshable Braille Display (Braille Terminal).
- [0303]** Printers, such as, but not limited to, inkjet printers, laser printers, 3D printers, solid ink printers and plotters.
- [0304]** Audio and Video (AV) devices, such as, but not limited to, speakers, headphones, amplifiers and lights, which include lamps, strobes, DJ lighting, stage lighting, architectural lighting, special effect lighting, and lasers.
- [0305]** Other devices such as Digital to Analog Converter (DAC)
- [0306]** Input/Output Devices may further comprise, but not be limited to, touchscreens, networking device (e.g., devices disclosed in network **1462** sub-module), data storage device (non-volatile storage **1461**), facsimile (FAX), and graphics/sound cards.
- [0307]** All rights including copyrights in the code included herein are vested in and the property of the Applicant. The

Applicant retains and reserves all rights in the code included herein, and grants permission to reproduce the material only in connection with reproduction of the granted patent and for no other purpose.

V. Aspects

[0308] The following disclose various Aspects of the present disclosure. The various Aspects are not to be construed as patent claims unless the language of the Aspect appears as a patent claim. The Aspects describe various non-limiting embodiments of the present disclosure.

Aspect 1 is a method for copying and modifying an automated invoice payment process from a marketplace, said method comprising:

[0309] listing of necessary states significant for a data processing for enabling automated invoice payment, wherein the necessary states are configured to perform at least the following:

[0310] awaiting for a bank customer in whose name an invoice is issued to enter a service channel,

[0311] awaiting delivery of an invoice to customer, and

[0312] awaiting customer consent to pay;

[0313] determining data parameters used for designation of the necessary states, wherein the parameters relate to at least the following: service type, customer ID, and invoice amount;

[0314] assembling a data structure and a process (objects) corresponding to information about at least the necessary states significant for the process;

[0315] passing requests through a graphical diagram (graph) of the process upon occurrence of an event significant for the process,

[0316] wherein the process moves from one state to a next state,

[0317] wherein the event is associated with at least one of the following:

[0318] the customers presence in the service channel,

[0319] delivery of the invoice, or

[0320] confirmed customer consent to pay; and

[0321] publishing said graph with a description of the process on a publicly-accessible web page among a plurality of independent graphs, each graph accessed by an end-user by clicking on the link and copying into the end-user's web page and said graph configured for modification by the end-user for a customized automated invoice payment process and launching the customized automated invoice payment process from the end-users web page.

Aspect 2 is the method of Aspect 1, further comprising a node, wherein the node comprises at least one of a start node, execute node, or a final node within a network of nodes executing the process.

Aspect 3 is the method of Aspect 2, wherein the execute node executes a task upon the condition with the set of parameters are met by the event.

Aspect 4 is the method of Aspect 2, wherein the execute node terminates a task loop upon an expiration of a predefined amount of time the process has been running.

Aspect 5 is the method of Aspect 2, wherein each of the any nodes comprise a queue, counter, and at least one function applied to the object.

Aspect 6 is the method of Aspect 5, wherein the any node comprises at least one of a unique function, counter, and queue.

Aspect 7 is the method of Aspect 1, wherein the object perpetually transfers from a node to at least one other node until at least one of an event obligation is satisfied or a counter or time threshold is reached.

Aspect 8 is the method of Aspect 2, wherein each node is associated with a function executed manually via at least one of an API, a code, or another node.

Aspect 9 is the method of Aspect 2, wherein each node comprises at least a counter time (Ct) measuring time of an oldest object in the queue.

Aspect 10 is the method of Aspect 2, wherein each node comprises at least a counter number (Cn) measuring a number of objects in the queue.

Aspect 11 is the method of Aspect 1, further comprising the step of monitoring progress of the process by an indicator.

Aspect 12 is a method for copying and modifying an automated invoice payment process from a marketplace, said method comprising the steps of:

[0322] determining data parameters used for designation of the necessary states enabling an automated invoice payment process, wherein the parameters relate at least to the following: service type, customer ID, and invoice amount;

[0323] assembling a process corresponding to information about at least the necessary states significant for the process in a form of a graphical diagram (graph); and

[0324] moving, upon occurrence of an event significant for the process, from one state to a next state, wherein the event is associated with at least one of the following:

[0325] the customers presence in the service channel,

[0326] delivery of the invoice, or

[0327] confirmed customer consent to pay; and

[0328] publishing said graph with a description of the process on a publicly-accessible web page among a plurality of independent graphs, each graph accessed by an end-user by clicking on the link and copying into the end-users web page and said graph configured for modification by the end-user for a customized automated invoice payment process and launching the customized automated invoice payment process from the end-users web page.

Aspect 13 is the method of Aspect 12, wherein the publishing of processes onto the publicly-accessible marketplace is controlled.

Aspect 14 is the method of Aspect 13, wherein the marketplace control is a singular and centralized authority.

Aspect 15 is the method of Aspect 13, wherein the marketplace control is distributive.

Aspect 16 is the method of Aspect 12, wherein the description of publicly-accessible processes describe the generic business process or pro-business outcomes achievable.

Aspect 17 is the method of Aspect 12, wherein the plurality of publicly-accessible processes are in listed generic business process or pro-business outcomes achievable form.

Aspect 18 is the method of Aspect 17, wherein clicking on any one of the listed form directs the end-user to a dedicated URL page.

Aspect 19 is the method of Aspect 18, wherein the end user modifies/customizes the process on any one of the dedicated URL page or end-user page after copying and pasting the link.

Aspect 20 is a method for end-user customization of a task automation process from a task automation marketplace, said method comprising:

[0329] publishing a graph with a description of a task automation process on a publicly-accessible web page

among a plurality of independent graphs, each graph accessed by an end-user clicking on the link and copying into the end-users web page; and

[0330] configuring said graph for modification by the end-user for launching the customized process from the end-users web page.

VI. Claims

[0331] While the specification includes examples, the disclosure's scope is indicated by the following claims. Furthermore, while the specification has been described in language specific to structural features and/or methodological acts, the claims are not limited to the features or acts described above. Rather, the specific features and acts described above are disclosed as example for embodiments of the disclosure.

[0332] Insofar as the description above and the accompanying drawing disclose any additional subject matter that is not within the scope of the claims below, the disclosures are not dedicated to the public and the right to file one or more applications to claims such additional disclosures is reserved.

The following is claimed:

1. One or more non-transitory computer readable media comprising instructions which, when executed by one or more hardware processors, causes performance of operations comprising:

creating a process flow based on an indication of a data processing task;

determining, for the data processing task, a listing of a plurality of necessary states significant for the data processing task;

determining data parameters used for designation of the necessary states;

creating a data structure associated with the data processing task, the data structure comprising the data parameters used for designation of the necessary states;

creating a plurality of nodes for representing the data processing task, each of the plurality of nodes being associated with a particular state, of the plurality of necessary states, each node being configured to perform a function associated with the particular state and to pass the data structure to another node, of the plurality of nodes, upon completion of the function;

publishing a graph associated with the data processing task, the graph comprising a plurality of vertexes corresponding to the plurality of nodes, and a plurality of edges associated with transitions between the nodes, wherein publishing the graph comprises making the graph available through a portal accessible by a plurality of end users;

receiving, from a first user of the plurality of users, a request to modify the published graph;

creating a modified graph based on the published graph, the modified graph comprising a change to one or more of the vertexes of the graph; and

creating a modified process flow based on the modified graph.

2. The media of claim 1, the operations further comprising:

configuring a system associated with the first user to execute the modified process flow.

3. The media of claim 1, wherein modifying the published graph comprises:

modifying at least one vertex of the published graph; and
modifying at least one node based on the modified vertex.

4. The media of claim 1, wherein the listing of the plurality of necessary states significant for the data processing task comprises:

waiting for a bank customer in whose name an invoice is issued to enter a service channel,
waiting for delivery of an invoice to customer, and
waiting for customer consent to pay; and
wherein the data parameters used for designation of the necessary states comprise: a service type, a customer ID, and an invoice amount.

5. The media of claim 1, the operations further comprising limiting access to the published graph based on one or more user parameters associated with the published graph, such that a particular user, of the plurality of users, that does not meet the one or more user parameters is not permitted to access the published graph.

6. The media of claim 5, wherein the user parameters comprise at least one of the following:

a user type;
a number of employees in an entity associated with the user; and
an indication of a revenue earned by the entity associated with the user.

7. The media of claim 1, the operations further comprising:

setting a monetary price associated with access to the published graph;
receiving payment from the first user; and
permitting the first user to access the published graph, responsive to receiving the payment.

8. A method comprising:

creating a process flow based on an indication of a data processing task;
determining, for the data processing task, a listing of a plurality of necessary states significant for the data processing task;

determining data parameters used for designation of the necessary states;

creating a data structure associated with the data processing task, the data structure comprising the data parameters used for designation of the necessary states;

creating a plurality of nodes for representing the data processing task, each of the plurality of nodes being associated with a particular state, of the plurality of necessary states, each node being configured to perform a function associated with the particular state and to pass the data structure to another node, of the plurality of nodes, upon completion of the function;

publishing a graph associated with the data processing task, the graph comprising a plurality of vertexes corresponding to the plurality of nodes, and a plurality of edges associated with transitions between the nodes, wherein publishing the graph comprises making the graph available through a portal accessible by a plurality of end users;

receiving, from a first user of the plurality of users, a request to modify the published graph;

creating a modified graph based on the published graph, the modified graph comprising a change to one or more of the vertexes of the graph; and

creating a modified process flow based on the modified graph.

9. The method of claim 8, further comprising:

configuring a system associated with the first user to execute the modified process flow.

10. The method of claim 8, wherein modifying the published graph comprises:

modifying at least one vertex of the published graph; and
modifying at least one node based on the modified vertex.

11. The method of claim 8, wherein the listing of the plurality of necessary states significant for the data processing task comprises:

waiting for a bank customer in whose name an invoice is issued to enter a service channel,
waiting for delivery of an invoice to customer, and
waiting for customer consent to pay; and
wherein the data parameters used for designation of the necessary states comprise: a service type, a customer ID, and an invoice amount.

12. The method of claim 8, further comprising limiting access to the published graph based on one or more user parameters associated with the published graph, such that a particular user, of the plurality of users, that does not meet the one or more user parameters is not permitted to access the published graph.

13. The method of claim 12, wherein the user parameters comprise at least one of the following:

a user type;
a number of employees in an entity associated with the user; and
an indication of a revenue earned by the entity associated with the user.

14. The method of claim 8, further comprising:

setting a monetary price associated with access to the published graph;
receiving payment from the first user; and
permitting the first user to access the published graph, responsive to receiving the payment.

15. A system comprising:

at least one device including a hardware processor;
the system being configured to perform operations comprising:

creating a process flow based on an indication of a data processing task;

determining, for the data processing task, a listing of a plurality of necessary states significant for the data processing task;

determining data parameters used for designation of the necessary states;

creating a data structure associated with the data processing task, the data structure comprising the data parameters used for designation of the necessary states;

creating a plurality of nodes for representing the data processing task, each of the plurality of nodes being associated with a particular state, of the plurality of necessary states, each node being configured to perform a function associated with the particular state and to pass the data structure to another node, of the plurality of nodes, upon completion of the function;

publishing a graph associated with the data processing task, the graph comprising a plurality of vertexes corresponding to the plurality of nodes, and a plurality of edges associated with transitions between the nodes, wherein publishing the graph comprises making the graph available through a portal accessible by a plurality of end users;

receiving, from a first user of the plurality of users, a request to modify the published graph;
creating a modified graph based on the published graph, the modified graph comprising a change to one or more of the vertexes of the graph; and
creating a modified process flow based on the modified graph.

16. The system of claim **15**, the operations further comprising:

configuring a system associated with the first user to execute the modified process flow.

17. The system of claim **15**, wherein modifying the published graph comprises:

modifying at least one vertex of the published graph; and
modifying at least one node based on the modified vertex.

18. The system of claim **15**, wherein the listing of the plurality of necessary states significant for the data processing task comprises:

waiting for a bank customer in whose name an invoice is issued to enter a service channel,

waiting for delivery of an invoice to customer, and
waiting for customer consent to pay; and
wherein the data parameters used for designation of the necessary states comprise: a service type, a customer ID, and an invoice amount.

19. The system of claim **15**, the operations further comprising limiting access to the published graph based on one or more user parameters associated with the published graph, such that a particular user, of the plurality of users, that does not meet the one or more user parameters is not permitted to access the published graph.

20. The system of claim **15**, the operations further comprising:

setting a monetary price associated with access to the published graph;

receiving payment from the first user; and

permitting the first user to access the published graph, responsive to receiving the payment.

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